POLYTECHNIQUE MONTRÉAL
affiliée à l'Université de Montréal

A Lightweight Canonicalization Engine in C for Disciplined Convex Programming

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Mémoire présenté en vue de l'obtention du diplôme de *Maîtrise ès sciences appliquées*
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A Lightweight Canonicalization Engine in C for Disciplined Convex Programming

présenté par **William ZHANG**
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DEDICATION

To all the past, present and future CVXPY contributors

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RÉSUMÉ

Le résumé est un bref exposé du sujet traité, des objectifs visés, des hypothèses émises, des méthodes expérimentales utilisées et de l'analyse des résultats obtenus. On y présente également les principales conclusions de la recherche ainsi que ses applications éventuelles. En général, un résumé ne dépasse pas trois pages.

Le résumé doit donner une idée exacte du contenu du mémoire ou de la thèse. Ce ne peut pas être une simple énumération des parties du manuscrit. Le but est de présenter de façon précise et concise la nature, l'envergure de la recherche, les sujets traités, les questions de recherche ou les hypothèses soulevées, les méthodes utilisées, les principaux résultats ainsi que les conclusions retenues. Un résumé ne doit jamais comporter de références ou de figures.

ABSTRACT

Written in English, the abstract is a brief summary similar to the previous section (Résumé). However, this section is not a word for word translation of the abstract in French.

The abstract is a brief statement of the subject matter, objectives, research questions or hypotheses, experimental methods and analysis of results. It also presents the main research conclusions and their possible applications. In general, an abstract should not exceed three pages.

The abstract should provide an exact idea of the thesis or dissertation's contents and it cannot be a simple enumeration of the manuscript's parts. The goal is to precisely and concisely present the nature and scope of the research. An abstract should never include references or figures. If the thesis or the dissertation is in English, the résumé (French-language abstract) should come first followed by the abstract.

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LIST OF SYMBOLS AND ACRONYMS

AD	Algorithmic Differentiation
AML	Algebraic Modelling Language
AMPL	A Mathematical Programming Language
ASA	Affine-Solver-Affine
ASL	AMPL Solver Library
BLAS	Basic Linear Algebra Subprograms
CSC	Compressed Sparse Column
CSR	Compressed Sparse Row
DAG	Directed Acyclic Graph
DCP	Disciplined Convex Programming
DPP	disciplined parameterized programming
JVP	Jacobian-vector product
LP	Linear Program
MOI	MathOptInterface
MPS	Mathematical Programming System
NLP	Nonlinear Program
QP	Quadratic Program
SIMD	Single Instruction, Multiple Data
SOCP	Second-Order Cone Program
SDP	Semidefinite Program
VJP	vector-Jacobian product

CHAPTER 1 INTRODUCTION

Since the early days of linear programming, optimization practitioners have needed to translate mathematical formulations into data that solvers can process. At first, this meant encoding each coefficient of the problem matrices by hand [1]. Matrix generators were introduced in the late 1960s to automate this task: they compute the coefficients through explicit loops and write them out in solver-readable formats like MPS [2, 3]. Matrix generators were able to remove most of the manual labour, but a gap still remained. Modellers naturally think in terms of sets, indices, and algebraic expressions, while procedural code constructs the coefficient matrices entry by entry. This mismatch between how a model is expressed and how it is implemented made models difficult to verify, modify, and communicate with others [4]. Algebraic modelling languages such as `AMPL` [5], `AIMMS` [6], and `GAMS` [7] closed this gap by accepting problem descriptions that closely follow the mathematical formulation. The survey [8] provides a complete overview of the early history of modelling languages.

More recently, modelling languages embedded in general-purpose programming languages have gained popularity. These include `Pyomo` [9,10] in Python, `JuMP` [11] in Julia, `CasADi` [12] in C++, and `YALMIP` [13] in MATLAB. Other domain-specific modelling languages specialized for convex optimization [14] are built on disciplined convex programming (DCP), the composition rules under which the convexity of a problem can be verified automatically [15, 16]. These DCP-based languages include `CVX` [17] in MATLAB, `CVXR` [18] in R, `CVXPY` [19] in Python, and `Convex.jl` [20] in Julia.

In the meantime, modern general-purpose solvers [21–23] have mostly transitioned towards conic programming, which generalizes linear programming and can represent and solve nearly every convex program used in practice [24]. The translation task remains a responsibility of the modelling languages, which now target conic problem data instead of Linear Program (LP) coefficient matrices. In DCP-based languages, this translation process is called canonicalization [25]. While the time taken to solve an optimization problem is often dominated by the solver, the time taken to canonicalize (or compile) it can also be significant [26]. The main goal of this Master's thesis is to reduce the compilation time of `CVXPY`.

1.1 Problem statement

The standard form of a convex optimization problem is

$$\begin{array}{ll} \underset{\mathbf{x}\in\mathbb{R}^n}{\text{minimize}} & f_0(\mathbf{x};\boldsymbol{\theta}) \\ \text{subject to} & f_i(\mathbf{x};\boldsymbol{\theta}) \le 0, \quad i = 1,\dots,m_1 \\ & g_j(\mathbf{x};\boldsymbol{\theta}) = 0, \quad j = 1,\dots,m_2 \end{array} \tag{1.1}$$

where $\mathbf{x} \in \mathbb{R}^n$ is the optimization variable, the objective $f_0 : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}$ and inequality constraint functions $f_1, \dots, f_{m_1} : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}$ are convex, and the equality constraint functions $g_1, \dots, g_{m_2} : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}$ are affine [14]. The parameter $\boldsymbol{\theta} \in \mathbb{R}^p$ specifies data that can change between solves but is fixed and given when solving an instance of the problem. We call the parametrized problem (1.1) a problem family; for a fixed value of $\boldsymbol{\theta}$, we call it a problem instance.

Canonicalization converts the high-level description (1.1) to a quadratic conic form used by many general-purpose convex solvers

$$\begin{array}{ll} \underset{\mathbf{x}\in\mathbb{R}^n}{\text{minimize}} & \frac{1}{2}\mathbf{x}^\top \mathbf{P}(\boldsymbol{\theta})\mathbf{x} + \mathbf{c}(\boldsymbol{\theta})^\top \mathbf{x} \\ \text{subject to} & \mathbf{b}(\boldsymbol{\theta}) - \mathbf{A}(\boldsymbol{\theta})\mathbf{x} \in \mathcal{K} \end{array} \tag{1.2}$$

where the *problem data* are the symmetric positive semidefinite matrix $\mathbf{P}(\boldsymbol{\theta}) \in \mathbb{S}^n_+$, the matrix $\mathbf{A}(\boldsymbol{\theta}) \in \mathbb{R}^{m\times n}$, the vectors $\mathbf{b}(\boldsymbol{\theta}) \in \mathbb{R}^m$ and $\mathbf{c}(\boldsymbol{\theta}) \in \mathbb{R}^n$, and the non-empty closed convex cone $\mathcal{K} \subseteq \mathbb{R}^m$. Problem (1.2) generalizes both quadratic programs (QPs), when $\mathcal{K} = \mathbb{R}^m_+$ is the nonnegative orthant, and standard conic programs, when $\mathbf{P}(\boldsymbol{\theta}) = 0$.

Most of the cost of canonicalization lies in the final step, where symbolic expression trees are converted to numerical problem data. This step is often called matrix stuffing or the backend of canonicalization [27]. During the first solve of a problem family, `CVXPY` forms and caches a parameter tensor that maps the parameter vector to the problem data. On subsequent solves, new problem data can be obtained with a single matrix-vector product [26].

For large-scale problems, forming the parameter tensor can be prohibitively expensive, so `CVXPY` lets users opt out of it entirely. Users must thus choose between paying a large upfront cost on the first solve and repeating the full canonicalization, with nothing reused, at every solve. We review `CVXPY` and the frontend of canonicalization in §1.2, as well as the parameter tensor and matrix stuffing in §2.1.

The constraint matrix $\mathbf{A}$ is the Jacobian of the stacked constraint expressions and $\mathbf{P}$ is the Hessian of the objective, so matrix stuffing amounts to computing a sparse Jacobian and

Hessian. Classical methods build these matrices column by column with multiple passes of algorithmic differentiation (AD), using graph colouring to group columns that can be recovered in a single pass [28, 29]. These methods treat the expression as a blackbox: the sparsity pattern must first be detected and the columns grouped before a single entry of the matrix is obtained. We review the basics of computing derivatives in §1.3 and present sparse AD and graph colouring methods in §2.2.

A modelling language, by contrast, knows which variables enter each constraint. `AMPL` (A Mathematical Programming Language) is split into two components: its commercial translator writes the structure of the model to a `.nl` file [30], and the open source AMPL Solver Library (ASL) evaluates constraints and their derivatives from this file inside the solver process [31]. `Pyomo` also interfaces to solvers through `.nl` files, among other formats, though it generally writes them more slowly than the `AMPL` translator.

On the other hand, `JuMP` has no single standard form to target; it decouples modeller and solver through MathOptInterface (MOI), a typed in-memory interface [32]. The model is built constraint by constraint, and each constraint stores its coefficients directly, so extraction reduces to copying them into the format of the chosen solver. None of these approaches operates on expression trees: the AD tools record the expression as a flat tape of instructions, `AMPL` exchanges the model through a file, and `JuMP` stores lists of coefficients. In `CVXPY` the expression trees themselves are the input of the backend. We review both these modelling languages in §2.3.

Our approach. We make use of the fact that the sparsity patterns of $\mathbf{A}$ and $\mathbf{P}$ are fixed across a problem family, because the parameter $\boldsymbol{\theta}$ changes the values of the problem data but not their structure. We can therefore split matrix stuffing into a symbolic and a numeric phase. The symbolic phase runs once on the first solve, where it computes the sparsity patterns, allocates memory for all intermediate matrices, and caches other computations that remain valid across the problem family. The numeric phase then fills in new values at every solve. This separation between structure and values is inspired by the interfaces of Nonlinear Program (NLP) solvers such as `IPOPT` [33, 34].

The symbolic phase is much cheaper to form and cache than the parameter tensor, and the numeric phase is often similar in cost to applying the matrix-vector product. The problem data is also assembled entirely in memory rather than through a separate model file. Finally, we exploit structure within the sparsity patterns themselves, which often contain dense blocks of data. When possible, we store these blocks as dense matrices and compute with dense linear algebra kernels, rather than always relying on generic sparse formats and routines (we

provide background on dense and sparse matrix computations in §1.4).

1.2 Canonicalization

The problem family (1.1) is written in functional form, as objective and constraint functions of the variable and the parameters. In a modelling language, the optimization variables $\mathbf{x}_1, \ldots, \mathbf{x}_k$ are collected into a single vector $\mathbf{x} \in \mathbb{R}^n$, and likewise the parameters $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_l$ into $\boldsymbol{\theta} \in \mathbb{R}^p$ [14]. Users are not required to write constraints with a zero right-hand side as in (1.1). Modelling languages accept inequalities with a convex left-hand side and a concave right-hand side, and equalities with both sides affine. Converting to the form (1.1) is trivial and only requires moving every term to the left-hand side. Internally, each function is represented as an *expression tree*.

Modelling languages for convex optimization are built on top of a library of functions which can be used to describe problems. These functions are known as *atoms* in `CVXPY` because they are the fundamental building blocks of the modelling language [15]. Alongside its numerical definition, each atom declares useful attributes such as its curvature, monotonicity with respect to each argument, and the sign of its output. A selection of common atoms is shown in Table 1.1.

Table 1.1 A selection of `CVXPY` atoms with their DCP attributes: curvature, monotonicity in each argument, and sign of the output.

Atom	Definition	Curvature	Monotonicity	Sign
`sum`	$\sum_i x_i$	affine	non-decreasing	any
`abs`	$\lvert x \rvert$	convex	unknown	nonnegative
`square`	x^2	convex	unknown	nonnegative
`sqrt`	$\sqrt{x}$	concave	non-decreasing	nonnegative
`exp`	e^x	convex	non-decreasing	positive
`log`	$\log x$	concave	non-decreasing	any
`norm`	$\lVert x \rVert_2$	convex	unknown	nonnegative
`max`	$\max_i x_i$	convex	non-decreasing	any
`sum_largest`	sum of k largest entries	convex	non-decreasing	any

An expression is defined recursively as an atom evaluated at subexpressions, each of which is itself another expression. Such an expression is naturally stored as an expression tree, with atoms at its internal nodes and variables, parameters, and constants at its leaves. In Python, this tree can be formed automatically from user code through operator overloading. Most operations on an expression, from evaluation to verifying convexity to differentiation,

traverse the tree and combine the intermediate results at each node.

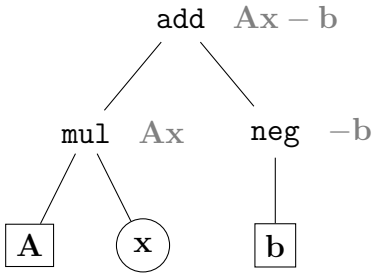


Figure 1.1 Example of an affine expression tree for $\mathbf{Ax} - \mathbf{b}$. Variables are drawn as circles and constants (or parameters) as boxes.

To certify the convexity of a problem, DCP-based modelling languages use the DCP composition rules to classify the curvature of each function [15]. Once all functions are classified, a problem is DCP-compliant if the objective and each constraint have curvatures compatible with a convex problem: a convex objective, convex $\leq$ concave inequalities, and affine equalities. Otherwise, if a curvature is incompatible (e.g., f_0 is concave) or cannot be determined, the modelling language raises an error and terminates. Because these rules are only sufficient for convexity, a problem that is genuinely convex but not expressed in a recognized form may still be rejected.

The composition $\phi(u_1(\mathbf{x}), u_2(\mathbf{x}), \ldots, u_k(\mathbf{x}))$ of an atom $\phi : \mathbb{R}^k \to \mathbb{R}$ with subexpressions $u_i : \mathbb{R}^n \to \mathbb{R}$ is convex when ϕ is convex and, for each argument $i \in \{1, \ldots, k\}$, one of the following holds:

- ϕ is non-decreasing in argument i and u_i is convex;
- ϕ is non-increasing in argument i and u_i is concave; or
- u_i is affine.

The same results hold for concave functions by swapping convex and concave. Affine expressions require ϕ and all of its arguments to be affine [15].

1.2.1 Conic reformulation

Once a problem is verified to be DCP-compliant, canonicalization rewrites its functional form (1.1) into an equivalent functional conic form

$$\begin{array}{ll} \underset{\mathbf{x}\in\mathbb{R}^n}{\text{minimize}} & f_0(\mathbf{x};\boldsymbol{\theta}) \\ \text{subject to} & h_i(\mathbf{x};\boldsymbol{\theta}) \in \mathcal{K}_i, \quad i = 1,\dots,m, \end{array} \tag{1.3}$$

where f_0 is affine or quadratic, each $h_i : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}^{d_i}$ is affine, and each $\mathcal{K}_i \subseteq \mathbb{R}^{d_i}$ is one of the standard cones supported by general-purpose conic solvers: the zero, nonnegative, second-order, power, exponential, and positive semidefinite cones.

Every nonlinear atom is rewritten into conic constraints in three steps. The first step rewrites the problem into *Smith form*, where each nonlinear atom is replaced by a new variable coupled with an equality constraint; every atom then acts only on affine arguments [35]. For a convex atom ϕ, the Smith form gives an auxiliary variable $t \in \mathbb{R}$ with the equality constraint $\phi(\mathbf{x}) = t$. The second step relaxes this equality to the inequality $\phi(\mathbf{x}) \le t$. The DCP rules allow a convex atom only where its value is minimized or upper-bounded, so the solver always prefers t as small as possible; the relaxation is therefore lossless. The pair $(\mathbf{x}, t)$ then lies in the *epigraph* $\{(\mathbf{x}, t) \in \mathbb{R}^n \times \mathbb{R} : \phi(\mathbf{x}) \le t\}$ of ϕ, so we often refer to t as an epigraph variable. A concave atom relaxes to $\phi(\mathbf{x}) \ge t$ instead, giving a hypograph variable. The third and final step expresses each epigraph (hypograph) inequality as the membership of an affine function of $(\mathbf{x}, t)$ in a convex cone [36].

Example of reformulations. A first example is the ℓ_2 norm atom $\|\mathbf{x}\|_2$ of $\mathbf{x} \in \mathbb{R}^n$, whose epigraph $\|\mathbf{x}\|_2 \le t$ is exactly the second-order cone constraint $(t, \mathbf{x}) \in \mathcal{K}_{\text{soc}}$, where $\mathcal{K}_{\text{soc}} = \{(t, \mathbf{x}) \in \mathbb{R} \times \mathbb{R}^n : \|\mathbf{x}\|_2 \le t\}$. A second example is the nondifferentiable absolute value atom $|x|$ of a scalar x, whose epigraph $|x| \le t$ can be written as the pair of affine inequalities $-t \le x \le t$. A final example is the sum of the k largest entries of $\mathbf{x}$, which expands into a small LP: the epigraph sum_largest$(\mathbf{x}, k) \le t$ holds if and only if there exist a scalar $q \in \mathbb{R}$ and a vector $\mathbf{w} \in \mathbb{R}^n$ such that

$$q\mathbf{1} + \mathbf{w} \ge \mathbf{x}, \qquad \mathbf{w} \ge 0, \qquad kq + \mathbf{1}^\top \mathbf{w} \le t, \tag{1.4}$$

where the inequalities are applied element-wise. Here q acts as a threshold and each w_i measures how far the entry x_i rises above it, so that the sum of the k largest entries is bounded by $kq + \mathbf{1}^\top \mathbf{w}$ [14].

1.2.2 `CVXPY`

`CVXPY` is a DCP-based modelling language embedded in Python [19], which began in 2013 as an implementation of the modelling principles of `CVX` [17]. It differed from `CVX` in three design decisions: modular components built around a few shared data structures, the expression tree formed by Python's operator overloading as the central representation, and support for parameters, which lets compilation work be reused when a problem is solved repeatedly. These ideas matured with version 1.0, which redesigned canonicalization as a general reduction-based rewriting system [25]. The new design enabled the addition of other disciplined grammars [34,37,38], and the early parameter support grew into DPP, a restricted grammar under which canonicalized programs can be updated efficiently [26].

The open source project has since grown through contributions from many other enthusiastic developers; a recent paper surveys the new canonicalization and modelling features introduced between versions 1.1 and 1.9 [27].

The rewriting from the functional form (1.1) to the functional conic form (1.3) is carried out as a sequence of *reductions*. Each reduction provides an `apply` step that converts a problem into an equivalent one, and a `retrieve` step that recovers a solution of the original problem from a solution of the rewritten one [25]. A first example is `FlipObjective`, which turns a maximization problem into a minimization one by flipping the sign of the objective, and flips it back in its `retrieve` step. A second example is `CvxAttr2Constr`, which replaces variable attributes such as nonnegativity with explicit constraints when the target solver does not support them natively.

The main reduction, `Dcp2Cone`, applies the epigraph reformulation of §1.2.1 in a bottom-up pass over the expression trees. The problem is first rewritten into Smith form, each resulting equality is relaxed to an epigraph (or hypograph) inequality, and each inequality is finally expressed as a cone membership [35, 36]. In practice the three steps are performed at once: each nonlinear atom is replaced directly by its epigraph variable and cone constraints, and the intermediate Smith form and its relaxation never appear explicitly.

1.2.3 Example

We consider the following simple fitting with regularization problem

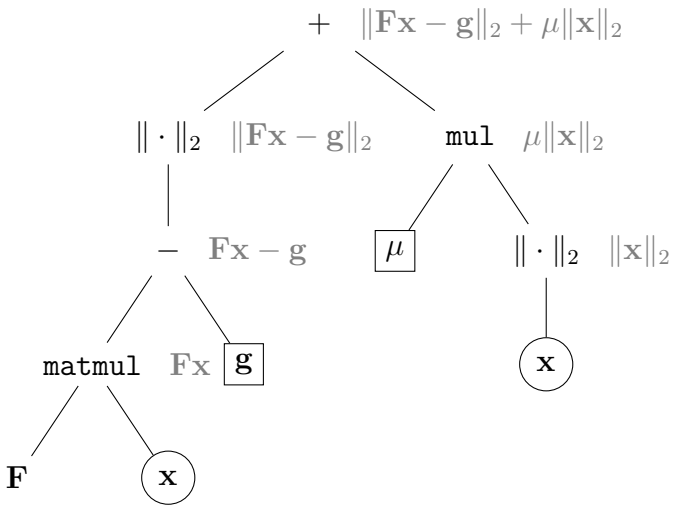
$$\begin{array}{ll} \underset{\mathbf{x}\in\mathbb{R}^n}{\text{minimize}} & \|\mathbf{F}\mathbf{x}-\mathbf{g}\|_2+\mu\|\mathbf{x}\|_2 \\ \text{subject to} & \mathbf{x}\geq 0 \end{array} \tag{1.5}$$

with variable $\mathbf{x} \in \mathbb{R}^n$, fixed problem data $\mathbf{F} \in \mathbb{R}^{m \times n}$, and parameters $\mathbf{g} \in \mathbb{R}^m$ and $\mu \geq 0$; the constraint $\mathbf{x} \geq 0$ requires every entry of $\mathbf{x}$ to be nonnegative. The problem formulation in `CVXPY` code is

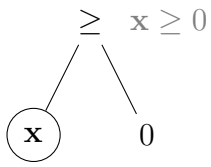
```
import cvxpy as cp
import numpy as np
F = np.random.randn(m, n)
x = cp.Variable(n)
g = cp.Parameter(m)
mu = cp.Parameter(nonneg=True)
objective = cp.Minimize(cp.norm(F @ x - g) + mu * cp.norm(x))
prob = cp.Problem(objective, [x >= 0])
```

Listing 1.1 The running example (1.5) in `CVXPY`.

where we also use the numerical computation library `NumPy` [39] for the data arrays. The expression tree representation of the problem is shown in Figure 1.2.



(a) Expression tree of the objective



(b) Expression tree of the constraints

Figure 1.2 Before canonicalization. `CVXPY` represents the DCP problem (1.5) as these expression trees. Variables are labelled with a circle, and parameters with a square.

After applying the reductions, the problem has been canonicalized to

$$\begin{array}{ll} \underset{t_1,\, t_2,\, \mathbf{x}}{\text{minimize}} & t_1 + \mu t_2 \\ \text{subject to} & \mathbf{x} \geq 0 \\ & (t_1, \mathbf{F}\mathbf{x} - \mathbf{g}) \in \mathcal{K}_{\text{soc},m+1} \\ & (t_2, \mathbf{x}) \in \mathcal{K}_{\text{soc},n+1} \end{array} \tag{1.6}$$

where $t_1, t_2 \in \mathbb{R}$ are new epigraph variables, and the cone $\mathcal{K}$ from (1.2) is the product $\mathbb{R}^n_+ \times \mathcal{K}_{\text{soc},m+1} \times \mathcal{K}_{\text{soc},n+1}$. Problem (1.6) is now in the functional conic form of §1.2.1: every constraint is a cone membership, and the objective and constraint expressions are affine. Comparing the DCP trees of Figure 1.2 with the trees of Figure 1.3 shows the effect of canonicalization: the nonlinear norm atoms have been replaced by epigraph variables, leaving affine expression trees in the variables, which we call the conic-form trees of the problem. The three rows of (1.6) remain three separate cone constraints, each carrying the affine expressions that must lie in its cone; Figure 1.3 accordingly shows one expression tree per constraint, rather than a single combined tree.

1.3 Computing derivatives

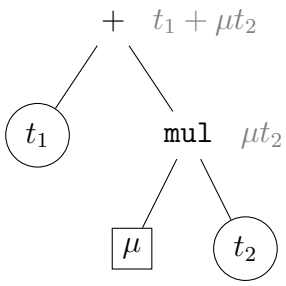
Computing derivatives is a fundamental task in scientific computing, with many methods that differ in cost, in accuracy, and in the form of the result [40]. The derivatives in this Master's thesis are the constraint Jacobian $\mathbf{A}$ and the objective Hessian $\mathbf{P}$ of a canonicalized problem, both large sparse matrices. This section presents the Jacobian and the Hessian, the elementary methods for computing them, and the choices that distinguish the software implementations; §2.2 reviews sparse AD and graph colouring methods.

For a vector-valued map $F : \mathbb{R}^n \to \mathbb{R}^m$ of an input $\mathbf{x} \in \mathbb{R}^n$, the first-order derivative is the Jacobian $\mathbf{J} \in \mathbb{R}^{m \times n}$ of partial derivatives of the outputs with respect to the inputs,

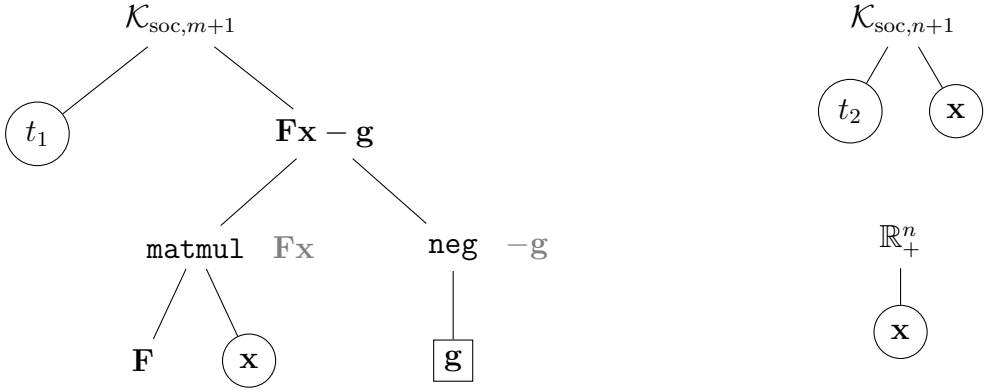
$$J_{ij} = \frac{\partial F_i}{\partial x_j}, \qquad i = 1, \ldots, m, \quad j = 1, \ldots, n. \tag{1.7}$$

We mainly consider the constraint function, which stacks the potentially many affine expressions of a canonicalized problem into a single vector function of the variables. Its Jacobian is the constant constraint matrix $\mathbf{A}$ of (1.2), whose sparsity pattern is fixed although its values may change from one solve to the next.

For a scalar function $f : \mathbb{R}^n \to \mathbb{R}$, the second-order derivative is the Hessian $\mathbf{H} \in \mathbb{R}^{n \times n}$ of



(a) Expression tree of the objective



(b) One expression tree per constraint

Figure 1.3 After canonicalization. `CVXPY` represents the conic form (1.6) as these expression trees: the nonlinear norm atoms have become epigraph variables, leaving the objective and the three separate cone constraints affine. The variable $\mathbf{x}$ appears as a leaf of several trees; all occurrences refer to the same variable.

second partial derivatives,

$$H_{ij} = \frac{\partial^2 f}{\partial x_i \, \partial x_j}, \qquad i, j = 1, \ldots, n. \tag{1.8}$$

The Hessian is symmetric whenever f is twice continuously differentiable, and it is then the Jacobian of the gradient of f. It can therefore be obtained by differentiating the gradient with the same methods used for the Jacobian, although dedicated methods that compute it directly also exist. The function we consider is the canonicalized objective, whose Hessian is the matrix $\mathbf{P}$ of (1.2).

1.3.1 Numerical differentiation

Other than computing derivatives analytically, the simplest method is to use numerical differentiation, which approximates the derivative of a function at a point by evaluating the function at nearby points using a finite difference formula such as

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}, \tag{1.9}$$

where h is a small step size. Note that the central variant $f'(x) \approx (f(x+h) - f(x-h))/2h$ improves accuracy. Both formulas use two function evaluations, but the forward difference can reuse the already computed value $f(x)$ across directions, while the central variant cannot. The approach is simple but has two downsides. First, it requires a linear number of function evaluations with respect to the number of input variables, which is expensive for high-dimensional functions. Second, it can easily lead to truncation errors, when h is too large, or rounding errors, when h is too small, in the computed derivative if the step size is not carefully chosen [41]. Despite these downsides, numerical differentiation is still widely used in practice, especially for verifying the correctness of derivative computations, also known as a derivative checker [42].

1.3.2 Symbolic differentiation

Mathematical expressions can be manipulated automatically using simple derivative rules of calculus, such as the product and sum rule. This symbolic approach to differentiation is used in computer algebra systems such as `Mathematica` [43] and `Maple` [44] as well as the `SymPy` library in Python [45]. Unlike its numerical counterpart, symbolic differentiation tools compute derivatives as closed-form expressions, which can be simplified and evaluated at any arbitrary point. These closed-form expressions are particularly useful for deriving

analytic solutions of optimization problems, and can be further used in mathematical proofs and analyses.

However, a naive implementation of symbolic differentiation can quickly form exponentially large symbolic expressions, which is impractical and inefficient for many problems [46]. When intermediate expressions are stored as common subexpressions, and the expression is represented with a directed acyclic graph (DAG) instead of a tree, symbolic differentiation performs the same set of operations as reverse mode differentiation [47].

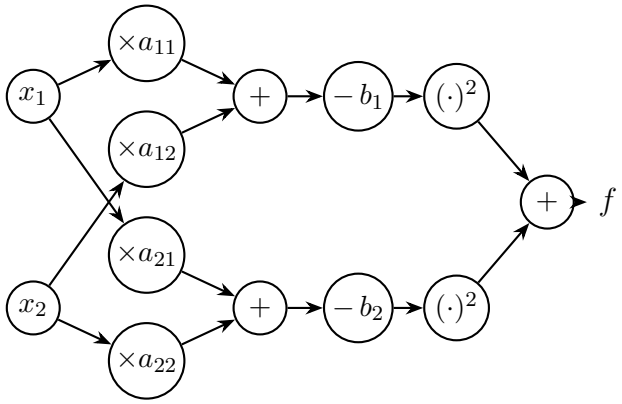
1.3.3 Algorithmic differentiation

Algorithmic differentiation (AD) treats the function as a computational graph of elementary operations, each with a known local derivative rule. The chain rule is then applied to combine these local derivatives throughout the graph, giving the derivative of the whole function. AD is also commonly referred to as automatic differentiation. We prefer the term algorithmic to emphasize that AD differentiates the algorithm that computes a function, rather than a numerical approximation or a symbolic formula [46].

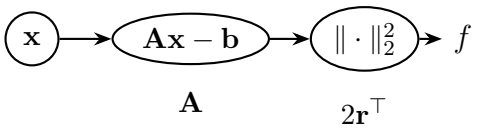
In this broad sense, AD can refer to any method that applies the chain rule over a computational graph. Its implementations differ in four basic choices: whether the graph is scalar or vectorized, whether the chain rule runs forward or in reverse, whether the derivative is assembled directly or accessed through matrix-vector products, and whether the result is stored sparse or dense.

In most AD tools, every node is a single elementary operation, an addition, a multiplication, or a univariate function, and every edge carries a scalar partial derivative [46]. In a vectorized graph, a node is a whole vector or matrix operation, and its local derivative is a matrix, a block of the Jacobian. Figure 1.4 contrasts the two on a small least squares objective. `CasADi` [12] offers both, through its scalar SX and vectorized MX expressions.

The modes of AD differ in the order in which the local derivatives are accumulated. Forward mode (or tangent mode) propagates derivatives from the inputs toward the outputs, following the order of evaluation. Reverse mode (or adjoint mode) propagates them from the outputs back toward the inputs, and requires storing the intermediate values of the evaluation [40]. In forward mode the work scales with the number of input variables, and in reverse mode with the number of output variables.



(a) Scalar graph: one node per elementary operation



(b) Vectorized graph: one node per matrix operation

Figure 1.4 The least squares objective $f(\mathbf{x}) = \|\mathbf{Ax} - \mathbf{b}\|_2^2$ with $\mathbf{A} \in \mathbb{R}^{2\times 2}$ as a scalar and as a vectorized computational graph. In the scalar graph every edge carries a scalar partial derivative. In the vectorized graph the local derivative of each node is a Jacobian block, shown below the node: $\mathbf{A}$ for the affine node and $2\mathbf{r}^\top$ for the sum of squares, where $\mathbf{r} = \mathbf{Ax} - \mathbf{b}$.

1.4 Matrix computations

The derivatives of §1.3 are matrices, and every matrix is stored in either a dense representation, which keeps all entries in a single array, or a sparse representation, which keeps only the nonzero entries alongside their positions. The representation determines the memory a matrix occupies and the algorithms that operate on it, and each has its own use cases. This section reviews the storage formats, algorithms, and some of the history behind both types of matrices.

1.4.1 Dense matrices

A dense matrix is stored as a single contiguous array of its entries, ordered row by row (row-major) or column by column (column-major), as shown in Figure 1.5. The position of an entry is computed directly from its indices, so no index arrays are needed, and rows or columns are read sequentially in memory. C arrays are row-major and Fortran arrays are column-major. Most dense linear algebra interfaces follow the Fortran convention.

Dense linear algebra is standardized by the Basic Linear Algebra Subprograms (BLAS), an interface of vector and matrix kernels [48, 49]. The BLAS is a specification rather than an implementation. Implementations such as `OpenBLAS`, Intel `MKL`, and Apple `Accelerate` are specialized for their respective processors, and nearly every machine has one available, whether shipped with the operating system or bundled by numerical libraries. LAPACK, the

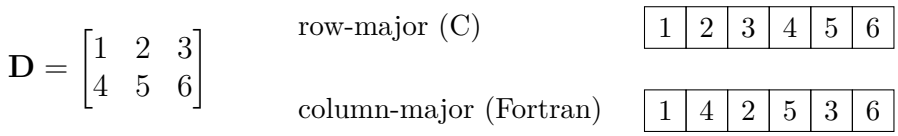


Figure 1.5 Dense storage of a 2×3 matrix: the same entries as one contiguous array in row-major and in column-major order.

library behind dense factorizations and linear system solving in most scientific software, is built on top of the BLAS kernels [50].

The BLAS interface is organized in three levels by the ratio of computation to data. Level 1 covers vector operations such as dot products, scaled copies, and additions, level 2 covers matrix-vector products, and level 3 covers matrix-matrix products. The routine names abbreviate the precision and the operation. For example, `dgemv` is the double-precision general matrix-vector product, and `dgemm` is the double-precision general matrix-matrix product.

BLAS implementations are fast because they are organized around the memory hierarchy. A modern processor multiplies numbers far faster than it loads them from main memory, so optimized kernels reuse data already in cache and process several entries per instruction with Single Instruction, Multiple Data (SIMD) units [51]. Level-3 kernels benefit the most because they perform a cubic number of operations on a quadratic amount of data, so each entry loaded from memory is reused many times.

1.4.2 Sparse matrices

In optimization problems, the matrices $\mathbf{P}$ and $\mathbf{A}$ often contain mostly zero entries. Such matrices are called sparse, and conic solvers expect their problem data in sparse formats. Techniques that store and compute with only the nonzero entries date back to the linear programming computations of the 1950s and the work of Markowitz [52]. The simplest sparse matrix data structure is the coordinate or triplet format. It consists of two integer index arrays `rows` and `cols` and a floating point value array `vals` of length equal to the number of nonzero entries in the matrix. For example, the matrix

$$\mathbf{S} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 3 & 4 \\ 0 & 5 & 6 & 0 \end{bmatrix} \tag{1.10}$$

can be represented in coordinate format as

```
int    rows[] = {0, 1, 1,  1,  2,  2};
int    cols[] = {1, 0, 2,  3,  1,  2};
double vals[] = {1, 2, 3, 4,  5,  6};
```

Listing 1.2 The matrix (1.10) in coordinate format.

where we use zero-based indexing and array declaration notation from the C programming language [53]. The coordinate format is the easiest to understand and form, but it is not effective for most operations on sparse matrices. The compressed sparse row (CSR) format is commonly used instead.

The CSR format differs from the coordinate format in that it replaces the `rows` array with a row pointer array `idxptr` of length equal to the number of rows plus one. The row pointer array indicates the starting index of each row in the `cols` arrays. For example, the matrix in (1.10) can be represented in CSR format as

```
int  idxptr[] = {0, 1,        4,     6};
int    cols[] = {1, 0, 2,  3,  1,  2};
double vals[] = {1, 2, 3, 4,  5,  6};
```

Listing 1.3 The matrix (1.10) in CSR format.

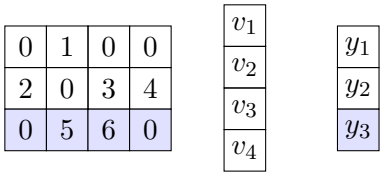
where we deliberately format `idxptr` to align with the start of each row in the `cols` arrays. The benefits of the CSR format are more than a reduced size of the `idxptr` array compared to the `rows` array. In the CSR format, accessing a row takes constant time and is very cache-friendly because the nonzero entries are stored contiguously in memory. For this reason, the CSR format is the default for iterative methods to solve sparse linear systems [54].

The compressed sparse column (CSC) format applies the same compression to the columns instead. The pointer array `idxptr` now points into a `rows` array, and the nonzero entries are stored column by column. The CSC arrays of a matrix read exactly as the CSR arrays of its transpose, so transposition is a conversion between the two formats, computable without sorting in one counting pass over the nonzeros [55]. Accessing a column of a CSR matrix instead requires a scan of every row, so algorithms that read columns prefer CSC; direct methods for sparse linear systems are the standard example [56].

The coordinate, CSR and CSC formats are the most widely used sparse matrix formats, but many others exist for different specialized use cases. The scientific Python sparse library `scipy.sparse` [57] allows users to form other types of sparse matrices such as the list of lists (LIL) matrix, the dictionary of keys (DOK) matrix, and the block-sparse row (BSR) matrix.

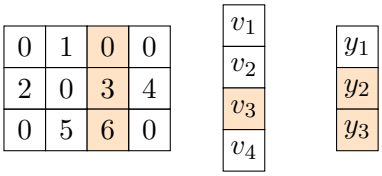
1.4.3 Sparse matrix products

Both compressed formats multiply a dense vector in one pass over the nonzeros, but the loop must follow the storage. CSR gathers, computing each output entry as a sparse dot product along its row, while CSC scatters, accumulating each column's contribution into the output vector. Figure 1.6 traces one product of the matrix in (1.10) under both formats.



$y_3 = 5v_2 + 6v_3$

(a) CSR gathers along a row



$y_2 \mathrel{+}= 3v_3, \quad y_3 \mathrel{+}= 6v_3$

(b) CSC scatters a column

Figure 1.6 The product $\mathbf{y} = \mathbf{S}\mathbf{v}$ with the matrix of (1.10). In CSR, the nonzeros of one row form a dot product that produces one output entry. In CSC, the nonzeros of one column are scaled by one input entry and accumulated into several output entries.

The product of two sparse matrices is formed row by row with Gustavson's algorithm [55]. Each row of the result is a combination of rows of the second operand, weighted by the nonzeros of the corresponding row of the first. Because the sparsity pattern of the result is not known in advance, the row is summed in a dense work array and then written back to sparse form [58]. It is the standard sparse matrix product, implemented in popular packages such as `Eigen` [59] and `scipy.sparse`.

In contrast to the BLAS kernels of §1.4.1, sparse products access memory irregularly. The loops run over the index arrays, so every nonzero costs an index load and an indirect access. Contiguous reads, SIMD, and cache tiling do not apply. Sparse formats are also a poor fit for dense data. A dense block stored in coordinate format triples its storage, one index pair per entry, and multiplying it through the sparse loop is far slower than the equivalent BLAS call. A sparse BLAS standard was specified by the BLAS Technical Forum [60], but implementations are few and far less widely adopted than the dense BLAS.

1.5 Contributions

We make the following contributions:

1. We build `SparseDiffEngine`, a C library with Python bindings that assembles sparse

Jacobians and Hessians directly from the expression structure, with no sparsity detection or graph colouring.

2. We implement a canonicalization backend for `CVXPY` built on `SparseDiffEngine`; this backend avoids the large upfront cost of the parameter tensor while still caching computations for subsequent solves.

3. We benchmark `SparseDiffEngine` against `CVXPY`'s existing canonicalization backends.

All code contributions are open source and available at

`https://github.com/SparseDifferentiation/SparseDiffengine`
`https://github.com/SparseDifferentiation/Jacobian-accumulation`

The rest of this thesis is organized as follows. Chapter 2 reviews how `CVXPY` and other modelling languages compute the problem data. In Chapter 3 we present an overview of the main data structures and algorithms in `SparseDiffEngine` as well as its interface in `CVXPY`. Finally, Chapter 4 benchmarks this engine against `CVXPY`'s three canonicalization backends, and finally, we conclude with a summary of our contributions and future improvements in Chapter 5.

CHAPTER 2 LITERATURE REVIEW

This chapter reviews the implementation of `CVXPY`'s matrix stuffing (§2.1) and of other modelling language approaches, from sparse AD such as `CasADi` (§2.2) to coefficient extraction in `AMPL` and `JuMP` (§2.3). These modelling languages all produce the same output as `CVXPY`, the solver problem data, but differ in the inputs they accept and in how they compute the coefficients. For example, `CasADi` accepts vectorized expressions as input, but differentiates them through matrix-vector products, so the structure of the derivatives is never materialized and must be detected at runtime, over many sweeps of the expression graph. `AMPL` and `JuMP` let the user declare vectorized expressions but scalarize them when processing the data into coefficients. The canonicalization backends of `CVXPY`, in contrast, keep the expressions vectorized and need no detection step (for this reason, we only compare against these backends in Chapter 4).

2.1 Matrix stuffing in `CVXPY`

Matrix stuffing converts the conic-form expression trees into the problem data. After canonicalization every expression is affine, so the Jacobian of each node is a constant matrix, which matrix calculus [61] gives in closed form. Computing the problem data reduces to composing these constant Jacobians over the tree. This section reviews the derivative rules along with two design choices of the `CVXPY` implementation: coefficient storage during the pass, and caching of the problem data across parameter changes. Finally, we describe `CVXPY`'s current backends and their limitations.

Matrix calculus extends differentiation to functions whose arguments and values are vectors and matrices. Systematic rules for such derivatives date back to the 1940s [62]. The simplest convention is to vectorize [61]: every expression is identified with

$$\operatorname{vec}(\mathbf{X}) = \begin{bmatrix} x_{11} & \cdots & x_{m1} & x_{12} & \cdots & x_{mn} \end{bmatrix}^\top, \tag{2.1}$$

the vector that stacks the columns of $\mathbf{X} \in \mathbb{R}^{m \times n}$, and the derivative of a vectorized expression with respect to another is an ordinary Jacobian matrix. Under this convention, matrix operations become linear operators on the vectorized arguments, and their Jacobians are structured. For example, the matrix product $\mathbf{FX}$ has the Kronecker-structured, block-diagonal Jacobian $\mathbf{I} \otimes \mathbf{F}$, shown in Figure 2.1.

$$\texttt{X = cp.Variable((2, 2))} \quad \Longrightarrow \quad \frac{\partial\, \mathrm{vec}(\mathbf{X})}{\partial\, \mathrm{vec}(\mathbf{X})} = \begin{array}{c} \\ x_{11} \\ x_{21} \\ x_{12} \\ x_{22} \end{array} \begin{array}{c} \begin{array}{cccc} x_{11} & x_{21} & x_{12} & x_{22} \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

(a) A matrix variable is the identity map of its own entries.

$$\texttt{Y = F @ X} \quad \Longrightarrow \quad \frac{\partial\, \mathrm{vec}(\mathbf{Y})}{\partial\, \mathrm{vec}(\mathbf{X})} = \mathbf{I} \otimes \mathbf{F} = \begin{array}{c} \\ y_{11} \\ y_{21} \\ y_{12} \\ y_{22} \end{array} \begin{array}{c} \begin{array}{cccc} x_{11} & x_{21} & x_{12} & x_{22} \end{array} \\ \begin{bmatrix} f_{11} & f_{12} & 0 & 0 \\ f_{21} & f_{22} & 0 & 0 \\ 0 & 0 & f_{11} & f_{12} \\ 0 & 0 & f_{21} & f_{22} \end{bmatrix} \end{array}$$

(b) The Jacobian of the product $\mathbf{Y} = \mathbf{FX}$ is block-diagonal, one copy of $\mathbf{F}$ per column of $\mathbf{X}$.

Figure 2.1 Jacobians under the vectorization (2.1), for a matrix variable $\mathbf{X}$ and the product $\mathbf{FX}$, with $\mathbf{X}, \mathbf{F} \in \mathbb{R}^{2\times 2}$. The entries of $\mathrm{vec}(\mathbf{X})$, in column-major order, index the columns of the Jacobians.

2.1.1 Composing linear operators

Every node of a conic-form tree, not only the root, computes an affine function of the problem variables and can be written in the vectorized form

$$f(\mathbf{z}) = \mathbf{A}_f\, \mathbf{z} + \mathbf{b}_f = \begin{bmatrix} \mathbf{A}_f & \mathbf{b}_f \end{bmatrix} \begin{bmatrix} \mathbf{z} \\ 1 \end{bmatrix}, \tag{2.2}$$

where $\mathbf{z}$ is the stacked vector of all the problem variables, $\mathbf{A}_f$ is the Jacobian of f with respect to $\mathbf{z}$, and $\mathbf{b}_f$ is a constant vector. Stacking the augmented matrices of the constraint trees yields the problem data $[\mathbf{A}\ \mathbf{b}]$ of (1.2). Matrix stuffing computes the augmented matrix in one bottom-up pass over the tree. A variable leaf contributes an identity block and a constant leaf contributes its value to the last column, as shown in Figure 2.3. An interior node with a single argument applies a constant linear operator $\mathbf{L}$ to the argument, which has the affine form (2.2) with augmented matrix $[\mathbf{A}_1\ \mathbf{b}_1]$; applying $\mathbf{L}$ gives

$$\begin{bmatrix} \mathbf{A}_f & \mathbf{b}_f \end{bmatrix} = \mathbf{L} \begin{bmatrix} \mathbf{A}_1 & \mathbf{b}_1 \end{bmatrix}. \tag{2.3}$$

The operator $\mathbf{L}$ is the Jacobian of the node with respect to its argument, so (2.3) is exactly the chain rule of matrix calculus. A node with k arguments applies one operator per argument,

$f = \sum_{i=1}^{k} \mathbf{L}_i f_i$, which gives

$$\begin{bmatrix} \mathbf{A}_f & \mathbf{b}_f \end{bmatrix} = \sum_{i=1}^{k} \mathbf{L}_i \begin{bmatrix} \mathbf{A}_i & \mathbf{b}_i \end{bmatrix}, \tag{2.4}$$

so the node costs one matrix product per argument.

In `CVXPY`, the linear operators divide into views and copies [63]. A view performs a simple manipulation of the coefficients of its argument. Atoms with one constant argument, such as `matmul` and `multiply`, make copies instead. The operator is a constant matrix built from the constant argument, and its product with the augmented matrix of the other argument is stored as a new coefficient matrix. Table 2.1 classifies a subset of the operators.

Table 2.1 A subset of the linear operators, classified as views or copies, applied to an argument f_1 with augmented matrix $[\mathbf{A}_1 \ \mathbf{b}_1]$, where $\otimes$ denotes the Kronecker product, $\odot$ the elementwise product, / the elementwise division, $\mathcal{I}$ an index set, and $\operatorname{diag}(\mathbf{v})$ the diagonal matrix with the entries of $\mathbf{v}$ on its diagonal.

Atom	**Operator L**	**Action on $[\mathbf{A}_1 \ \mathbf{b}_1]$**
Views		
$-f_1$	$-\mathbf{I}$	flip the sign of every stored entry
$f_1^\top$	$\mathbf{K}$, the vec-permutation	permute the rows
$f_1[\mathcal{I}]$	$\mathbf{P}_{\mathcal{I}}$, a row selection	keep the rows indexed by $\mathcal{I}$
$\operatorname{reshape}(f_1)$	$\mathbf{I}$	none, the vectorization is unchanged
Copies		
$\mathbf{F} f_1$	$\mathbf{I} \otimes \mathbf{F}$	multiply by a block diagonal formed from $\mathbf{F}$
$f_1 \mathbf{B}$	$\mathbf{B}^\top \otimes \mathbf{I}$	multiply by a Kronecker matrix formed from $\mathbf{B}$
$\mathbf{C} \odot f_1$	$\operatorname{diag}(\operatorname{vec}(\mathbf{C}))$	multiply by a diagonal formed from $\mathbf{C}$
$f_1 / \mathbf{C}$	$\operatorname{diag}(\operatorname{vec}(\mathbf{C}))^{-1}$	multiply by the reciprocal diagonal

2.1.2 Example

We continue the example of §1.2.3, which canonicalized problem (1.5) into the conic-form trees of Figure 1.3. We now compute the problem data of these trees. The variables are first stacked into $\mathbf{z} = (t_1, t_2, \mathbf{x})$. Each variable then occupies a contiguous block of columns of $\mathbf{A}_f$, starting at its column offset within $\mathbf{z}$. The columns of the variables that do not appear in the expression f are zero, because f does not depend on their entries. Figure 2.2 gives an example of column offsets on another set of variables.

Matrix stuffing then runs bottom-up over the objective and the three constraint trees, as shown in Figure 2.3. Each leaf of $\mathbf{x}$ contributes an identity block of size n at the column offset of $\mathbf{x}$, and the constant leaf $\mathbf{g}$ contributes its value to the last column. The interior nodes

$$\mathbf{z}^\top = \begin{array}{c} \quad\mathbf{u}\quad\quad\quad\mathbf{v}\quad\quad\quad\quad\mathbf{w} \\ \begin{array}{cccccccc} 0 & & 2 & & & 5 & & \\ [u_1 & u_2 & v_1 & v_2 & v_3 & w_1 & w_2] \end{array} \end{array}$$

Figure 2.2 The column offsets of a stack of three variables $\mathbf{z} = (\mathbf{u}, \mathbf{v}, \mathbf{w})$ with $\mathbf{u} \in \mathbb{R}^2$, $\mathbf{v} \in \mathbb{R}^3$, and $\mathbf{w} \in \mathbb{R}^2$. Each offset is the total dimension of the preceding variables: $\mathbf{u}$ starts at column 0, $\mathbf{v}$ at 2, and $\mathbf{w}$ at 5. The entries of $\mathbf{z}$ index the columns of every augmented matrix $[\mathbf{A}_f\ \mathbf{b}_f]$.

apply their operators to the matrices of their children; the product node $\mathbf{Fx}$, for example, multiplies the identity block of its argument by $\mathbf{F}$. Finally, the root stacks the augmented matrices of its children at their constraint row offsets into $[\mathbf{A}\ \mathbf{b}]$.

The augmented matrices of (2.2) can be stored in two ways, shown in Figure 2.4. In the global layout, drawn in Figure 2.3, each node stores one matrix whose columns span the whole stacked vector $\mathbf{z}$, with the column offsets applied from the start. In the local layout, which `CVXPY` uses [63], each node keeps one coefficient block per variable in its subtree, and one for the constant data. The offsets are applied only after the pass, when each block is placed at its variable's columns and the blocks are concatenated into the full matrix $[\mathbf{A}\ \mathbf{b}]$. The local layout stores only the columns of the variables present in a subtree. The tradeoff is that each interior node applies its operator once per stored block, rather than once per argument.

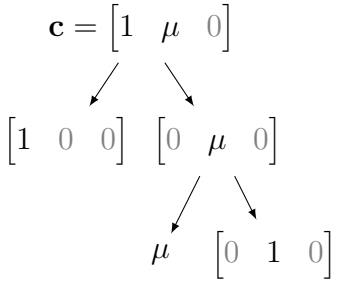
2.1.3 Parameter caching

In a problem family the structure is fixed and only the numeric parameter values change between solves. Under additional rules on how the parameters enter the expressions, known as disciplined parameterized programming (DPP), the problem data is affine in the parameters [26]. To keep track of the parameter dependence, `CVXPY` adds a third dimension to the problem data, indexed by the parameter entries. Every coefficient matrix throughout the bottom-up pass becomes a tensor. At the root nodes we thus get the cached parameter maps

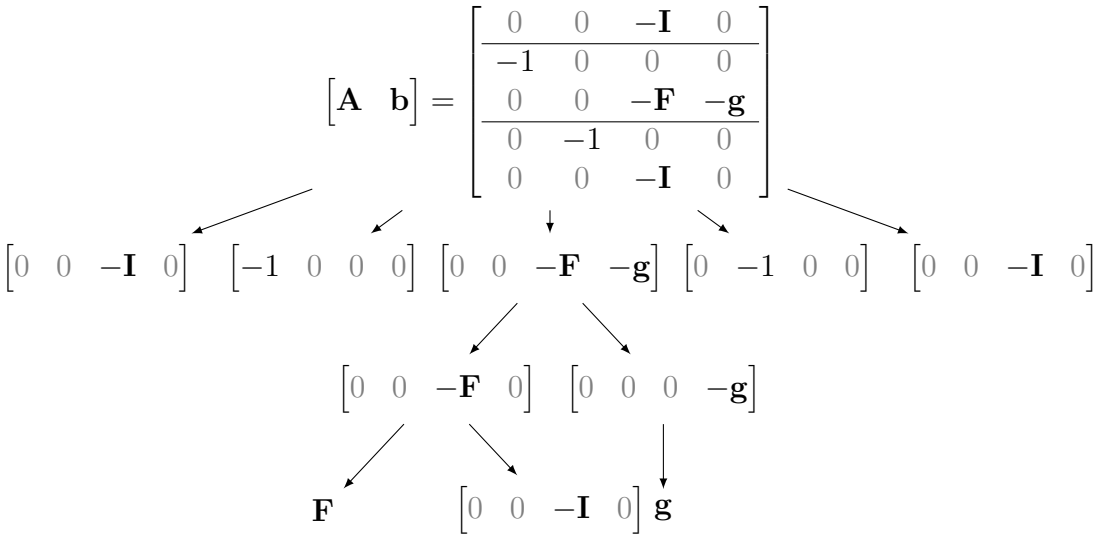
$$\mathbf{c} = \mathbf{Q}\,\tilde{\boldsymbol{\theta}}, \qquad \operatorname{vec}([\mathbf{A}\ \mathbf{b}]) = \mathbf{R}\,\tilde{\boldsymbol{\theta}}, \qquad \tilde{\boldsymbol{\theta}} = \begin{bmatrix} \boldsymbol{\theta} \\ 1 \end{bmatrix}, \tag{2.5}$$

where $\mathbf{Q} \in \mathbb{R}^{n\times(p+1)}$ and $\mathbf{R} \in \mathbb{R}^{m(n+1)\times(p+1)}$. The objective map $\mathbf{Q}$ is shown in Figure 2.5 for (1.5). After a parameter change, a re-solve costs one sparse matrix-vector product per map.

Problems that do not conform to the DPP grammar, or that opt out of it with the option `ignore_dpp`, take a different approach to matrix stuffing. The reduction `EvalParams` replaces



(a) Matrix stuffing of the objective



(b) Matrix stuffing tree of the constraints

Figure 2.3 Matrix stuffing of (1.5) by a bottom-up pass over the conic-form expression trees of Figure 1.3: each node's matrix is obtained by applying a linear operation to the matrices of its children. The coefficients appear negated because (1.2) writes the constraints as $\mathbf{b} - \mathbf{A}\mathbf{x} \in \mathcal{K}$. The horizontal rules in $[\mathbf{A}\ \mathbf{b}]$ separate the rows of the three constraints.

$$\left\{ \begin{array}{cc} \mathbf{x} & 1 \\ \mathbf{x} \mapsto [-\mathbf{F}], & -1 \mapsto [-\mathbf{g}] \end{array} \right\}$$

(a) Local layout.

$$\begin{array}{cccc} t_1 & t_2 & \mathbf{x} & 1 \\ [0 & 0 & -\mathbf{F} & -\mathbf{g}] \end{array}$$

(b) Global layout.

Figure 2.4 What each layout stores during the pass, at the node $-(\mathbf{F}\mathbf{x} - \mathbf{g})$ of (1.5): a dictionary of narrow per-variable blocks, or a single matrix spanning all columns.

$$\mathbf{Q} = \begin{array}{c} \\ t_1 \\ t_2 \\ \mathbf{x} \end{array} \begin{array}{c} \mathbf{g}^\top \\ \begin{bmatrix} 0 \cdots 0 \\ 0 \cdots 0 \\ 0 \cdots 0 \end{bmatrix} \end{array} \begin{array}{c} \mu \\ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \end{array} \begin{array}{c} 1 \\ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \end{array} \qquad \mathbf{c} = \mathbf{Q}\,\tilde{\boldsymbol{\theta}} = \begin{bmatrix} 1 \\ \mu \\ 0 \end{bmatrix}$$

Figure 2.5 The DPP path on the objective data of (1.5): the parameter map $\mathbf{Q}$ is built and cached once, and each re-solve recovers $\mathbf{c}$ with the single sparse product $\mathbf{Q}\tilde{\boldsymbol{\theta}}$.

each parameter leaf by a constant holding its current value, and the resulting parameter-free tree goes through the pass as if the problem had no parameters, as shown in Figure 2.6. The result is the same objective data $\mathbf{c}$ as in Figure 2.5, now obtained by repeating the whole matrix stuffing instead of applying the cached product $\mathbf{Q}\tilde{\boldsymbol{\theta}}$. The `ignore_dpp` path is more general, as it supports arbitrary functions of the parameters.

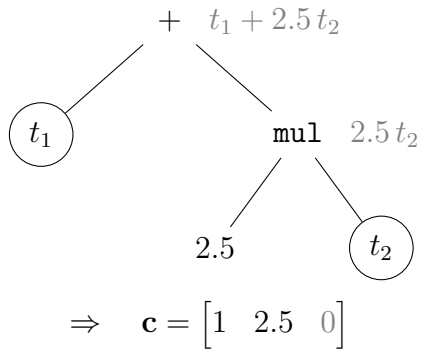


Figure 2.6 The `ignore_dpp` path on the objective of (1.5): the parameter leaf μ is replaced by a constant holding its current value 2.5, and the pass is repeated on the parameter-free tree on every change of the parameters.

`CVXPY` implements the pass in three canonicalization backends that differ in how they represent the parameter axis [64]:

- the C++ backend `cvxcore` keeps one sparse matrix per parameter slice and loops over the slices at every operator;
- the `SciPy` backend stacks the slices vertically into a single strided two-dimensional sparse matrix;
- the `COO` backend does not form the tensor explicitly, and instead stores each nonzero with an explicit parameter index alongside its row and column.

When canonicalization remains too slow, `CVXPYgen` can compile a DPP problem ahead of time [65]. It generates C code specific to the problem family, with the entries of the cached

maps (2.5) written into the source as constants, so a re-solve applies the parameter maps in compiled code.

The pass of this section produces the affine data $\mathbf{c}$ and $[\mathbf{A}\ \mathbf{b}]$. For a quadratic objective, `CVXPY` replaces the affine argument of each quadratic atom by a new variable, adding the corresponding equality constraint; the objective becomes a sum of quadratic forms with constant matrices. A second matrix stuffing pass then assembles these matrices into the diagonal blocks of the matrix $\mathbf{P}$ of (1.2) [63].

All three backends share two limitations. Firstly, they delegate their matrix operations to a general-purpose sparse library, `Eigen` [59] in `cvxcore` and `scipy.sparse` [57] in the Python backends. The storage formats and kernels are fixed by the library, so structure known at canonicalization time cannot be exploited. In `cvxcore`, the views of Table 2.1 also materialize explicit sparse operators. For example, a reshape builds a sparse identity matrix and multiplies the coefficients by it. Secondly, the canonicalization backends store dense constant matrices as sparse matrices. When dense blocks appear in the problem data, they are both stored in an inconvenient format and computed without the optimized BLAS kernels of §1.4.1. Our approach in Chapter 3 addresses these limitations.

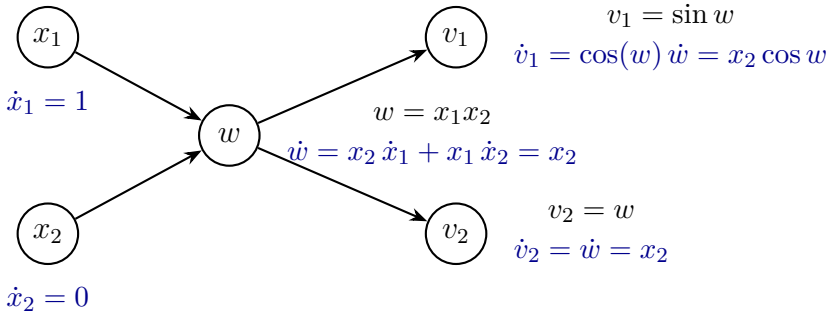
2.2 Sparse derivatives from detected structure

This section reviews methods that compute the full sparse Jacobian of a vector-valued function, given as an arbitrary program in a general-purpose language. No modelling language declares the structure of the program, so the sparsity of the derivatives must be detected [66]. Modern surveys of AD address dense derivatives, or the gradient of a scalar function, which reverse mode computes in a single sweep over the computational graph [40, 67]. In those settings, AD refers to the evaluation of matrix-vector products with the Jacobian or the Hessian, and the explicit matrix is never formed. Factorizations, direct linear solves, and many optimization solvers instead require the explicit matrix [66]. The methods of this section recover the full sparse matrix from multiple matrix-vector products [46]. To reduce the number of products, graph colouring groups the vectors that can be propagated together, so that one product recovers several columns at once.

2.2.1 Matrix-vector products and accumulation

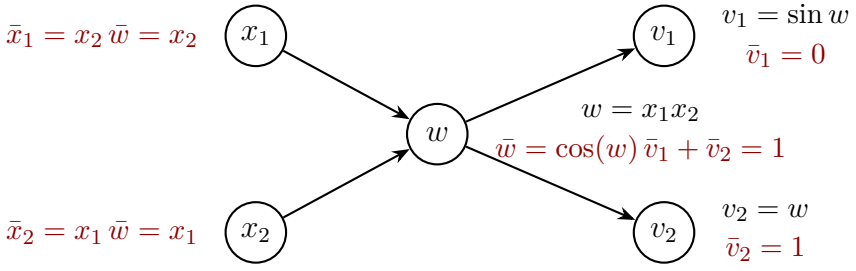
AD evaluates derivatives by sweeps of the computational graph. A sweep visits every node once and applies the chain rule at each, at about the cost of one function evaluation [46]. Each sweep computes one matrix-vector product with the derivative. Forward mode starts from a

seed vector $\mathbf{v}$ at the inputs and propagates tangents, and the Jacobian-vector product (JVP) $\mathbf{Jv}$ appears at the outputs. Reverse mode starts from a seed $\mathbf{w}$ at the outputs and propagates adjoints, and the vector-Jacobian product (VJP) $\mathbf{J}^\top\mathbf{w}$ appears at the inputs, as shown in Figures 2.7 and 2.8. For the Hessian, the analogous Hessian-vector product $\mathbf{Hv}$ comes from one forward sweep over one reverse sweep.



$$\begin{bmatrix}\dot{v}_1\\ \dot{v}_2\end{bmatrix} = \begin{bmatrix}x_2\cos w\\ x_2\end{bmatrix} = \mathbf{J}\mathbf{e}_1, \qquad \mathbf{J} = \begin{bmatrix}x_2\cos w & x_1\cos w\\ x_2 & x_1\end{bmatrix}.$$

Figure 2.7 Forward sweep of the computational graph of $F(x_1, x_2) = (\sin(x_1x_2),\ x_1x_2)$, with the intermediate value $w = x_1x_2$. Alongside its value, each node carries one tangent $\dot{v}$ (in blue), updated by the chain rule following the order of evaluation. The seed $(\dot{x}_1, \dot{x}_2) = (1, 0)$ delivers the first column of $\mathbf{J}$ at the outputs.

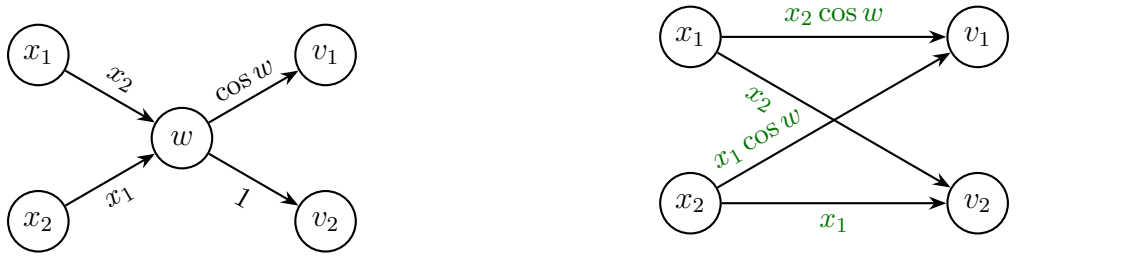


$$\begin{bmatrix}\bar{x}_1 & \bar{x}_2\end{bmatrix} = \begin{bmatrix}x_2 & x_1\end{bmatrix} = \mathbf{e}_2^\top\mathbf{J}.$$

Figure 2.8 Reverse sweep of the computational graph of Figure 2.7. Each node carries one adjoint $\bar{v}$ (in red), updated by the chain rule against the order of evaluation. The seed $(\bar{v}_1, \bar{v}_2) = (0, 1)$ delivers the second row of $\mathbf{J}$ at the inputs.

Recovering the full matrix takes a sequence of sweeps, one per seed vector. The n unit seed vectors recover $\mathbf{J}$ column by column in forward mode, and the m unit seed vectors recover it row by row in reverse mode [46]. The other approach is accumulation [68], which combines

the local derivatives of the graph directly into **J** by eliminating one intermediate vertex at a time, as shown in Figure 2.9. Finding the optimal elimination order is NP-complete [46, 69]. Batching the n unit seeds into one forward sweep gives each node a whole tangent, the derivative of that node with respect to all the inputs. Eliminating the vertices in the order of evaluation stores the same derivative at each node, so the batched sweep and forward-order elimination perform the same computation.



(a) Each edge carries a local derivative.

(b) After eliminating w, the edges are the entries of **J**.

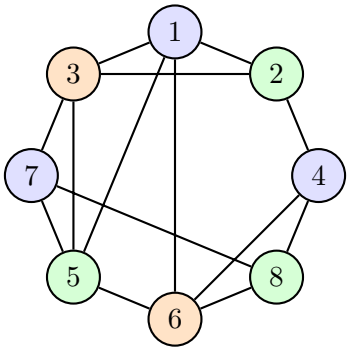
Figure 2.9 Accumulation of the Jacobian by vertex elimination, on the graph of Figure 2.7. Eliminating the intermediate vertex w connects each of its predecessors to each of its successors and multiplies the local derivatives along the path (in green); parallel edges, when they arise, are summed. Once every intermediate vertex is eliminated, the graph is bipartite and its edges are the entries of **J**.

In practice, AD tools such as `ADOL-C` [70] and `CppAD` [71] record the scalar graph as a runtime trace, one record per elementary operation [46, 72]. A single matrix-vector product unrolls into one scalar multiplication for every entry of the matrix, together with the additions that sum them, so the trace of a realistic problem can sometimes reach millions of nodes. On such a trace, every intermediate vertex would store its derivative with respect to all the inputs, so accumulation is rarely used in practice. When **J** is sparse, most of each sweep is then wasted on zeros, and the cost of recovering the matrix scales with its dimension rather than with its number of nonzeros.

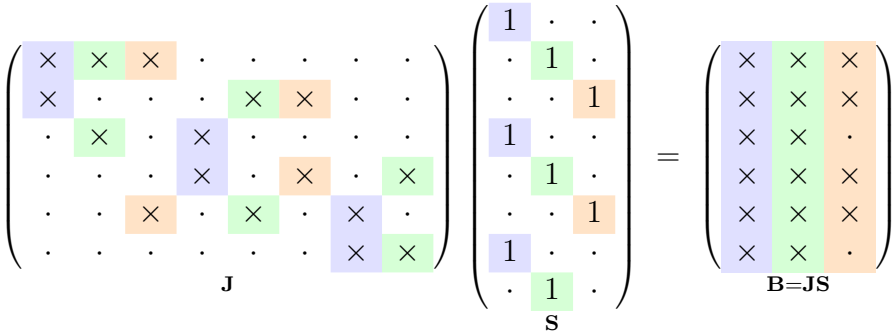
2.2.2 Compression by graph colouring

Several columns of a sparse Jacobian can share one seed [28]. Two columns with no nonzero in a common row are structurally orthogonal, and one sweep recovers both, each entry landing in a distinct row of the product. Finding the fewest such groups was later formalized as a colouring of the column intersection graph, which has one vertex per column and an edge between two columns that share a row [29, 73]. Figure 2.10 colours the graph of a 6×8 Jacobian and recovers the matrix in three sweeps instead of eight. The approach extends

to the Hessian, which compresses further because its symmetry allows dedicated star and acyclic colourings [74, 75].



(a) The column intersection graph of **J**, with one vertex per column and an edge between two columns that share a row, coloured with three colours.



(b) The seed matrix **S** sums the unit vectors of each colour class, and the compressed product **B** = **JS** recovers every nonzero of **J**, each in a distinct row of its compressed column.

Figure 2.10 Graph colouring compresses the seeds. Colouring the column intersection graph groups the columns of the sparse Jacobian **J** into three structurally orthogonal classes, and the matrix is recovered in three sweeps instead of eight.

The sparse Jacobian is obtained from the products in three stages: detect the sparsity pattern, compress it by colouring, and propagate the compressed seeds of Figure 2.10 through the graph to recover the nonzero entries [66]. Detection propagates sparsity patterns instead of values through the same graph, replacing each arithmetic operation by a union of index sets [46]. The pattern and its colouring are computed once and reused, so their cost is spread across repeated evaluations of the same function.

2.2.3 CasADi

CasADi [12] is an open-source framework for nonlinear optimization and optimal control built around AD. It is widely used in academia and industry, notably for model predictive control, trajectory optimization, robotics, and energy systems [76]. Its core is written in C++ and is used from Python, MATLAB, or Octave. It provides the building blocks of gradient-based optimization: symbolic expressions, derivatives, integrators, and interfaces to NLP and QP solvers. The gradients, Jacobians, and Hessians these solvers require are obtained by differentiating the model.

CasADi supports two types of expressions, and both return the matrix derivatives in a sparse format, which is expected by most solvers [76]. An SX graph is scalar, with exact node-by-node sparsity, but its size grows with the problem data. An MX graph is vectorized. Its nodes are whole matrix operations, so the graph stays small even when the model is large. Either graph can be exported as self-contained C code and compiled ahead of time for embedded applications.

CasADi computes its derivatives by the detection, colouring, and batched sweeps of this section, and caches the pattern, the colouring, and the compiled derivative function across evaluations. Listing 2.1 rebuilds the conic-form trees of Figure 1.3, obtained after canonicalization, as MX expressions; differentiating them recovers the problem data that matrix stuffing assembles in §2.1.2.

```
import casadi as cs
t1, t2 = cs.MX.sym("t1"), cs.MX.sym("t2")
x = cs.MX.sym("x", n)
g, mu = cs.MX.sym("g", m), cs.MX.sym("mu")
z = cs.vertcat(t1, t2, x)               # stacked variables
expr = cs.vertcat(x, t1, F @ x - g, t2, x) # stacked affine expressions
A = -cs.jacobian(expr, z)               # the matrix A
b = cs.substitute(expr, z, 0)           # the vector b
c = cs.gradient(t1 + mu * t2, z)        # the objective vector c
```

Listing 2.1 The conic form of the running example in CasADi.

The Jacobian with respect to the stacked variable $\mathbf{z}$ recovers the matrix $\mathbf{A}$ of Figure 2.3 entry by entry; differentiation is the equivalent of matrix stuffing, with the coefficients obtained by those sweeps rather than by a bottom-up pass over the tree. The parameters remain symbols, so $\mathbf{b}$ and $\mathbf{c}$ are expressions of $\mathbf{g}$ and μ, re-evaluated for each parameter value. This is how the qpsol interface builds its problem data internally, for problems with a quadratic

objective and affine constraints. A model written at the same high level as in `CVXPY`, with the norms of (1.5) kept in the expression graph, is solved through `nlpsol` instead. The same sweeps assemble the gradient, constraint Jacobian, and Hessian of the Lagrangian that the NLP solver queries at each iteration.

2.3 Coefficient extraction from declared structure

In contrast to §2.2, the methods of this section require the user to declare the structure of the problem. In `AMPL` and `JuMP`, the user writes each objective and constraint as a sum of terms, and the coefficient of every linear and quadratic term is stored as it is typed. Neither language has an atom library or a canonicalization step, so a nonlinear expression such as a norm is left to the solver (`AMPL`) or written in conic form by the user (`JuMP`). In place of differentiation, `AMPL` writes the stored coefficients to a `.nl` file that the solver parses, and `JuMP` keeps them as typed in-memory term lists that are copied to the solver.

2.3.1 `AMPL`

`AMPL` [5] is a commercial algebraic modelling language for large-scale mathematical programming, developed at Bell Laboratories in the 1980s. It is one of the most widely used modelling languages in academia and industry, applied to production planning, scheduling, logistics, and energy systems. A model is written declaratively, close to its mathematical statement, from sets, indexed parameters, indexed variables, and algebraic expressions over them. Data is supplied separately, so one model can be used for many problem instances. Models are written in the `AMPL` language itself or built through its Python interface [77], and reach a large collection of commercial and open-source solvers through a single instance format.

The translator runs as its own process. It parses the model and the data, instantiates every constraint, and writes the resulting instance to a `.nl` file [30]. Each solver is reached through a driver, a separate executable that reads the `.nl` file and calls the solver. A driver is built on one of two libraries that supply the reader and the expression handling. The AMPL Solver Library (ASL) is the original, a C library that parses the `.nl` format and evaluates the expressions it holds [31]. `MP` is the newer one, which AMPL supplies for solvers that need no derivatives [78]. This division between a translator that writes an instance and a driver that reads it still defines how most solvers receive nonlinear problems.

The `.nl` file is an instance format rather than a model format. Sets and indexing are resolved by the translator, and what remains is one flat record per objective and constraint. An indexed declaration such as `var x {1..n}` becomes n scalar variables, and an indexed

constraint becomes one record per index, so the file has no notion of a vector. The linear part of each record is an explicit list of (variable, coefficient) pairs, stored alongside the bounds and the nonzero counts of the instance. Only the nonlinear parts are kept as expression graphs, written in prefix notation [30].

The language has no norm atom and no canonicalization step, so the norms of problem (1.5) are entered as square roots of sums of squares. What the solver receives depends on its driver. An `MP`-based driver rewrites the two norms into second-order cones for a conic solver, and an ASL-based driver passes on a general nonlinear program.

```
param m; param n;
param F {1..m, 1..n};
param g {1..m};
param mu >= 0;
var x {1..n} >= 0;
minimize obj:
    sqrt(sum {i in 1..m} (sum {j in 1..n} F[i,j]*x[j] - g[i])^2)
  + mu * sqrt(sum {j in 1..n} x[j]^2);
```

Listing 2.2 The running example (1.5) in `AMPL`.

For an instance with $m = 3$, $n = 2$, and $\mu = 0.1$, the translator writes the following `.nl` file. The comments are added by hand to make the records readable and are not written by the translator.

```
g3 1 1 0    # text format; 9 more header lines follow
 2 0 1 0 0  # 2 vars, 0 constraints, 1 objective, 0 ranges, 0 eqns
...
O0 0        # objective 0, an expression graph in prefix notation
o0          #   add
o39         #     sqrt
o54         #       sum of 3 terms, one per row of F
...         #         affine subtrees F[i,1]*x[1] + F[i,2]*x[2] - g[i]
o2          #   multiply
n0.1        #     the value of mu
...         #     sqrt subtree of the second norm
b           # variable bounds: x[1] >= 0, x[2] >= 0
2 0
2 0
```

Listing 2.3 An abbreviated `.nl` file for an instance of the running example.

The whole objective is written as one expression graph, the constraint $\mathbf{x} \geq 0$ reduces to the two bound records of the `b` segment, and the value of μ is folded into the graph as the constant `n0.1`, so a parameter change requires a new file. For a linear or quadratic program, no graph remains and the file contains the complete problem data as coefficient lists.

Inside the driver, the bounds and the linear coefficient lists are read directly from the file, at a cost linear in the number of nonzeros, and only the nonlinear parts are rebuilt as expression graphs. What happens to those graphs depends on the library. `MP` matches second-order and exponential cones in the rebuilt graphs. A conic solver such as Gurobi or MOSEK then receives cone constraints rather than the square roots the model states. ASL instead differentiates the graphs by reverse-mode AD [79], with sparse Hessians assembled from partial separability [80].

Any system that writes `.nl` reaches every solver that reads it. The format has grown since, with extensions such as logical constraints, and AMPL omits the added header fields when a model does not use them, so older drivers still read the files it writes [30]. An instance is an ordinary file, so it can be archived, shipped to reproduce a bug, and rerun on another machine. However, serializing the instance to disk can be slow, and every solve repeats it, rebuilding any expression graphs from scratch.

2.3.2 `JuMP`

`JuMP` [11, 81] is an open-source algebraic modelling language embedded in Julia, developed at MIT and first released in 2013. It is the most widely used modelling language in Julia, applied in academia and industry to power systems, portfolio optimization, scheduling, and routing. A model is written with the macros `@variable`, `@constraint`, and `@objective`, and the same model can be passed to any of more than fifty supported solvers [82]. The abstraction layer between models and solvers is MOI [32].

The macros of `JuMP` rewrite each expression as it is entered. `@objective` and `@constraint` build affine and quadratic expression objects term by term, so the model stores coefficient lists instead of an expression tree. These lists are typed by MOI. A `VectorAffineFunction` is a list of (row, coefficient, variable) terms plus a constant vector. A `ScalarQuadraticFunction` adds a list of (coefficient, variable, variable) triples. A constraint states that one such function lies in a set. For a nonnegative orthant or a second-order cone, this is exactly the functional conic form of §1.2.1. MOI also supports other sets, such as integrality, special ordered sets, and indicator constraints [83].

The user declares the epigraph variables and cone memberships of (1.6) directly.

```
using JuMP
model = Model()
@variable(model, x[1:n] >= 0)
@variable(model, t1); @variable(model, t2)
@constraint(model, [t1; F * x - g] in SecondOrderCone())
@constraint(model, [t2; x] in SecondOrderCone())
@objective(model, Min, t1 + mu * t2)
```

Listing 2.4 The running example (1.5) in `JuMP`.

The first cone constraint is stored as

```
VectorAffineFunction([(1, 1.0, t1), (2, F[1,1], x[1]), ...],
                     [0.0, -g[1], ..., -g[m]]) in  SecondOrderCone(m + 1)
```

Listing 2.5 The first cone constraint as stored by `MOI`.

and the expression itself is not retained after the macro returns.

When the model is passed to a solver, MOI copies each constraint into the solver's own representation with a `copy_to` function. Solvers that want matrix data commonly implement that copy through the `MatrixOfConstraints` utility, which concatenates the stored term lists into one sparse matrix in the solver's format [83]. Coefficient extraction passes once over the stored terms and forms the sparse matrix, with no expression graph to visit and no differentiation.

When a solver does not support a constraint type natively, MOI reformulates it through its bridge system. A bridge is a structural rewrite from one function-set pair to another, for instance from a rotated to a standard second-order cone. MOI selects and composes them automatically [32]. Bridging manipulates stored coefficients and adds auxiliary variables and constraints.

`JuMP` also has a nonlinear path, in which expression graphs are retained and derivatives are computed by sparse AD. The MOI representation is general, and a new problem class enters as a new function or set rather than as a change to the interface. However, vectorized expressions written in the model are stored as scalar coefficients, and each is expanded into one term per nonzero as the macro runs.

CHAPTER 3 A LIGHTWEIGHT CANONICALIZATION ENGINE

Mathematically, `SparseDiffEngine` computes the same derivatives as `CVXPY`'s canonicalization backends. Both apply the chain rule of matrix calculus to trees of vectorized expressions. The differences are in the implementation. Firstly, a custom sparse matrix data structure allows a natural split between a symbolic pass and a numeric pass. This separation is particularly useful for compiling problems involving parameters, where the structure is fixed but the numerical values can change. Secondly, the data of most optimization problems is not only sparse but also structured, with dense blocks and diagonal patterns. In `SparseDiffEngine` we introduce two new matrix classes for this purpose, `permuted-dense` and `stacked-permuted-dense`, which let us use dense storage and BLAS subroutines when possible.

Finally, some important design decisions differ from `CVXPY`'s backends for simplicity rather than performance. Every node carries one Jacobian taken with respect to all of the variables of the problem, rather than a dictionary of local Jacobians keyed by variable. As a consequence, the chain rule is a single matrix product, and the constraint matrix is built by stacking the Jacobians of the constraints. In addition, quadratic functions have a dedicated Hessian pass, instead of a variable substitution followed by a second matrix stuffing pass.

The rest of the chapter is organized as follows. We begin with a brief overview of the engine, describing how expressions are represented and how the code is organized. We then present the matrix representations used for the derivative computations, where we focus much of our attention on the `permuted-dense` matrix. Next we describe the algorithms, or kernels, which operate on and combine these matrix representations. Finally, we describe the problem interface of the engine, and how `CVXPY` obtains and updates the problem data through it.

3.1 Overview of the engine

`SparseDiffEngine` is a small C library which is around seventeen thousand lines of code (and a similar number of lines for the test suite). The expression node is defined in `expr.h`; atoms that carry extra data, such as the exponent of a power, define a larger structure in `subexpr.h`. Each atom is implemented in its own file in the `atoms/` directory; there are around forty-five of them, divided into six subdirectories based on their mathematical properties. The matrix data structures and the computational kernels that operate on them are in `utils/`, and `problem.c` collects the objective and constraints of a problem and evaluates the values and

derivatives that a solver asks for.

3.1.1 Atom classes

Atoms in the same subdirectory compute their derivatives in the same way (Table 3.1). Canonicalization leaves the constraints affine and the objective affine or quadratic. The quadratic atoms of a DCP problem (a squared power, a quadratic over a linear term, a Huber penalty) are all rewritten as quadratic forms by the canonicalization frontend (§2.1), so only affine atoms and quadratic forms reach the engine from a canonicalized DCP problem. This chapter describes the affine atoms. The quadratic form contributes the objective matrix $\mathbf{P}$ through the Hessian (§3.4). The remaining classes are used by the nonlinear programming (NLP) interface of `CVXPY` [34]. Each atom implements three operations: a forward pass,

Table 3.1 The atom classes and how each computes its Jacobian.

Class	Examples	Jacobian computation
affine	`sum`, `index`, `left_matmul`	fixed; index maps and copies
elementwise, full domain	`exp`, `power`, `logistic`	diagonal scaling of the child Jacobian
elementwise, restricted domain	`log`, `entr`, `atanh`	diagonal at the variable's columns
bivariate, full domain	`multiply`, `matmul`	two-term chain rule
bivariate, restricted domain	`quad_over_lin`, `rel_entr`	two-term chain rule
other	`quad_form`, `prod`	specific to each atom

which evaluates the atom at a point; a symbolic pass, which computes the sparsity patterns of its derivatives and allocates memory for them; and a numeric pass, which computes the numerical values inside the fixed sparsity patterns. The constructor of the atom stores pointers to these three functions in the expression node.

3.1.2 Expression node

An expression node stores its dimensions, pointers to at most two children, a buffer for its value, and its derivative matrices. Every derivative matrix has `n_vars` columns, the total number of entries of all the variables of the problem.

```
typedef struct expr
{
    int d1, d2, size, n_vars, refcount, var_id;
    struct expr *left;
    struct expr *right;

    double *value;   /* forward value              */
    matrix *jacobian; /* Jacobian w.r.t. the variables */

    forward_fn      forward;
    jacobian_init_fn jacobian_init_impl; /* symbolic pass */
    eval_jacobian_fn eval_jacobian;    /* numeric pass */
} expr;
```

Listing 3.1 A simplified version of the expression node in `expr.h`.

Trees are built bottom up, with one constructor function per atom. A `CVXPY` expression is converted in one pass over the tree, which calls the constructor of the corresponding atom at each node (§3.4). Expressions that appear multiple times can be shared, in which case the tree becomes a directed acyclic graph. Nodes are therefore reference counted, and the symbolic pass returns early if a node was already initialized.

Unlike `CVXPY`, there are only two leaf nodes, variables and parameters. A variable occupies a contiguous slice of the stacked variable vector, starting at the column offset `var_id`. Its Jacobian is an identity block at the columns of the variable, written once during the symbolic pass. The Jacobian of a parameter is empty, because parameters are fixed during a solve. A constant is a parameter whose value cannot be updated between solves.

3.2 Matrix representations

The Jacobian of a node is stored in one of three representations, a generic sparse matrix, a permuted dense matrix, or a stacked permuted dense matrix. The generic sparse matrix is the default. The other two are created by matrix multiplication and the quadratic form, the operations that hold dense data. Each implements the same operations, so an atom picks one when it allocates its Jacobian during the symbolic pass.

3.2.1 Sparse matrix

The generic sparse representation stores a Jacobian in the CSR format of §1.4.2. A Jacobian has one row for each entry of the value of its node, and most operations, such as indexing, broadcasting, and stacking, operate on those rows. Each row is stored contiguously and reached in constant time. For this reason we chose the CSR format over the CSC format. In `SparseDiffEngine` the column indices within each row are kept sorted. The CSR format does not necessarily require this order. `Eigen` keeps its indices sorted as well [59], while `scipy.sparse` does not always enforce it [57]. The merges and intersections of the symbolic pass depend on this sorted order guarantee.

The symbolic pass allocates the row pointer and column index arrays, and the numeric pass writes only the value array. We provide more details on the benefits of this split between the sparsity pattern and the values in §3.3.2.

3.2.2 Permuted dense matrix

The Jacobians that appear during canonicalization do not always have scattered, isolated nonzeros. Multiplying an expression by a data matrix, for example, places a dense block at a subset of the rows and columns of the Jacobian. Consider the expression $\mathbf{Bx}$ with dense data. Its Jacobian holds the data at the columns of the variable and is zero everywhere else (Figure 3.1). Those columns are usually a small part of the Jacobian. The generic sparse representation would instead need to store a column index for each entry of the block.

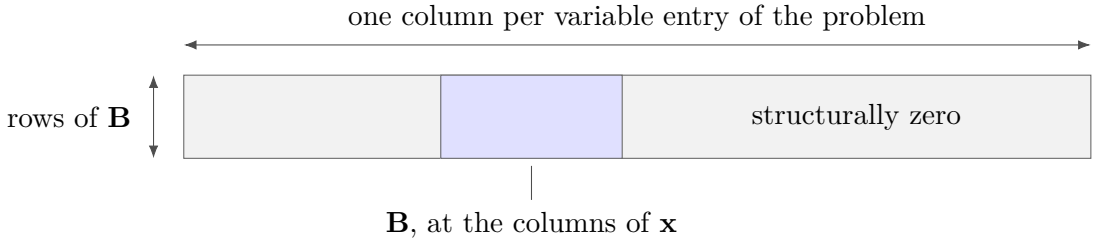


Figure 3.1 The Jacobian of $\mathbf{Bx}$ with respect to the full variable vector, for dense data $\mathbf{B}$. Its nonzeros are exactly the entries of $\mathbf{B}$.

The dense block can instead be stored together with the indices of the rows and columns it occupies, one index per row and column rather than one per entry, and the global matrix is never formed. A permuted dense matrix is defined as the triple $(\mathbf{X}, \mathbf{r}, \mathbf{c})$, with $\mathbf{X} \in \mathbb{R}^{m_0 \times n_0}$ dense and $\mathbf{r}$ and $\mathbf{c}$ strictly increasing index vectors, representing

$$\mathbf{A} = \mathbf{S}_r^\top \mathbf{X} \mathbf{S}_c \in \mathbb{R}^{m \times n}, \qquad A_{r_i, c_j} = X_{ij}, \tag{3.1}$$

where the rows of $\mathbf{S}_r \in \{0,1\}^{m_0 \times m}$ and $\mathbf{S}_c \in \{0,1\}^{n_0 \times n}$ are the rows of the identity indexed by $\mathbf{r}$ and $\mathbf{c}$ (Figure 3.2). Every other entry of $\mathbf{A}$ is structurally zero and never stored. The entries of $\mathbf{c}$ are columns of the global variable vector and the entries of $\mathbf{r}$ are rows of the Jacobian.

$$\mathbf{A} = \begin{pmatrix} \cdot & \cdot & \cdot & \cdot & b_{00} & b_{01} & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & b_{10} & b_{11} & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & b_{20} & b_{21} & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & b_{30} & b_{31} & \cdot & \cdot & \cdot & \cdot \end{pmatrix} \qquad \mathbf{X} = \begin{pmatrix} b_{00} & b_{01} \\ b_{10} & b_{11} \\ b_{20} & b_{21} \\ b_{30} & b_{31} \end{pmatrix}$$

$$\mathbf{r} = (0,1,2,3), \qquad \mathbf{c} = (4,5).$$

$$\mathbf{A} = \underbrace{\begin{pmatrix} 1 & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot \\ \cdot & \cdot & 1 & \cdot \\ \cdot & \cdot & \cdot & 1 \end{pmatrix}}_{\mathbf{S}_r^\top} \underbrace{\begin{pmatrix} b_{00} & b_{01} \\ b_{10} & b_{11} \\ b_{20} & b_{21} \\ b_{30} & b_{31} \end{pmatrix}}_{\mathbf{X}} \underbrace{\begin{pmatrix} \cdot & \cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot \end{pmatrix}}_{\mathbf{S}_c}$$

Figure 3.2 The Jacobian of $\mathbf{Bx}$ as a permuted dense matrix, for dense $\mathbf{B} \in \mathbb{R}^{4\times 2}$ and a variable $\mathbf{x} \in \mathbb{R}^2$ at columns $(4,5)$ of a problem with ten variable entries. Only the dense block $\mathbf{X}$ is stored, and the selectors are the rows of the 4×4 and 10×10 identities indexed by $\mathbf{r}$ and $\mathbf{c}$.

The dense block is stored in row-major order and the index vectors as two integer arrays, so the storage does not depend on the global dimensions m and n. The structural atoms turn a permuted dense matrix into another permuted dense matrix, and require only manipulating the row and column indices. Transposition swaps the two index vectors and transposes the dense block. Figure 3.3 indexes the first and last rows of the Jacobian of $\mathbf{Bx}$. The row vector of the result is $(0,1)$ rather than $(0,3)$, because it indexes the rows of the result. The map back to the rows of the original block is kept internally, and the numeric pass copies the values through this map at every evaluation.

A permuted dense matrix has nonzeros only inside its dense block. The Jacobian of $\mathbf{Bx} + \mathbf{y}$ contains the block of $\mathbf{B}$ at the columns of $\mathbf{x}$ and an identity at the columns of $\mathbf{y}$, which no single block covers. The sum is instead performed on CSR views of the two operands. A view is built during the symbolic pass, and its value array points to the dense block, so no values are copied. The result is stored as a generic sparse matrix, and so is everything computed from it.

$$\mathbf{A}[(0,3),\,:] \;=\; \begin{pmatrix} \cdot & \cdot & \cdot & \cdot & b_{00} & b_{01} & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & b_{30} & b_{31} & \cdot & \cdot & \cdot & \cdot \end{pmatrix}$$

$$\text{represented by the triple} \quad \left(\begin{pmatrix} b_{00} & b_{01} \\ b_{30} & b_{31} \end{pmatrix},\; (0,1),\; (4,5) \right).$$

Figure 3.3 Row indexing of the $\mathbf{A}$ of Figure 3.2. The result is mostly zeros, but its nonzeros still form one smaller dense block.

3.2.3 Stacked permuted dense matrix

Multiplying a matrix-valued expression with n rows and p columns by data $\mathbf{B} \in \mathbb{R}^{m\times n}$ multiplies each of the p columns by $\mathbf{B}$. On the vectorized expression the multiplication acts as the block diagonal operator $\mathbf{I}_p \otimes \mathbf{B}$ of Table 2.1. In practice this operator is not formed, unlike in `CVXPY`'s backends. The atom stores a single copy of $\mathbf{B}$ and applies it to one column at a time. Its Jacobian contains one dense block for each of the p columns, at pairwise disjoint rows and in general at different columns (Figure 3.4). No single triple $(\mathbf{X}, \mathbf{r}, \mathbf{c})$ describes such a matrix. Merging the blocks into one larger block would store the zeros between them, and falling back to the generic sparse representation would lose the dense structure. We instead keep the blocks as they are, in a third representation.

A stacked permuted dense matrix is defined as a vertical union of permuted dense blocks whose row vectors are pairwise disjoint, while the column vectors may overlap. Every operation treats the blocks one at a time, and the global matrix is never formed. The stacked matrix owns a single value buffer, and each block points into it.

$$\mathbf{M} \;=\; \begin{pmatrix} \cdot & \cdot & \cdot & \cdot & b_{00} & b_{01} & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & b_{10} & b_{11} & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & b_{00} & b_{01} & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & b_{10} & b_{11} & \cdot & \cdot \end{pmatrix}$$

$$\mathbf{X}_1 = \mathbf{X}_2 = \mathbf{B}, \qquad \mathbf{r}_1 = (0,1), \quad \mathbf{c}_1 = (4,5), \qquad \mathbf{r}_2 = (2,3), \quad \mathbf{c}_2 = (6,7).$$

Figure 3.4 The Jacobian of $\mathbf{BV}$ as a stacked permuted dense matrix, for dense $\mathbf{B} \in \mathbb{R}^{2\times 2}$ and a variable $\mathbf{V} \in \mathbb{R}^{2\times 2}$ at columns $(4,\ldots,7)$ of a problem with ten variable entries. There is one block per column of $\mathbf{V}$, and the 4×10 matrix $\mathbf{M}$ is never formed in practice.

3.3 Computational kernels

An atom does not operate on the representations of §3.2 directly. It instead calls a kernel, the implementation of one operation for one matrix representation. The ten operations of Table 3.2 cover the whole library, so every atom computes its derivatives through a small set of shared routines. Each kernel is a pair of functions, one called by the symbolic pass and one by the numeric pass.

Table 3.2 The kernels, the operation each computes, and the atoms of the conic path that call them, grouped into unary and binary operations.

Kernel	Computes	Called by
`transpose`	$\mathbf{A}^\top$	`left_matmul`
`index`	$\mathbf{A}[\mathbf{r}, :]$	`index`, `transpose`
`sum_row_partition`	sums of groups of rows	`sum`
`promote`	copies of a single row	`promote`
`broadcast`	copies of a row or a column	`broadcast`
`diag_vec`	rows moved to the diagonal positions	`diag_vec`
`DA`	$\mathrm{diag}(\mathbf{d})\,\mathbf{A}$	`vector_mult`, `quad_form`
`sum_matrices`	$\mathbf{A} + \mathbf{B}$	`add`, `quad_form`
`BA`	$\mathbf{BA}$	`left_matmul`, `quad_form`
`BTA`, `BTDA`	$\mathbf{B}^\top\mathbf{A}$, $\mathbf{B}^\top\mathbf{DA}$	`quad_form`

3.3.1 Kernel dispatch

The operations of Table 3.2 divide into unary operations, which operate on a single matrix, and binary operations, which combine two matrices. Dispatching a unary operation is simple. Each representation implements every unary operation, and a call on a Jacobian runs the corresponding implementation. The kernel of a binary operation is determined by the representations of both operands.

For example, on a generic sparse matrix the symbolic pass of the `index` kernel allocates a new CSR pattern and copies the column indices of the selected rows. On a permuted dense matrix it intersects the selected rows with the row index vector, as in Figure 3.3. On a stacked matrix the permuted dense kernel is applied to each block, and the blocks left with no selected rows are dropped. In all three representations the numeric pass only copies the values of the rows that were kept.

Of the three binary operations, only the transposed products dispatch on both operands. Addition always operates in the CSR format. Both operands are read through their CSR

views and the result is a generic sparse matrix, so no dispatch is needed. In the product $\mathbf{BA}$, the matrix $\mathbf{B}$ is data rather than a Jacobian, so only the representation of $\mathbf{A}$ selects the kernel. The kernel of the transposed products $\mathbf{B}^\top\mathbf{A}$ and $\mathbf{B}^\top\mathbf{DA}$ is selected by double dispatch (Figure 3.5).

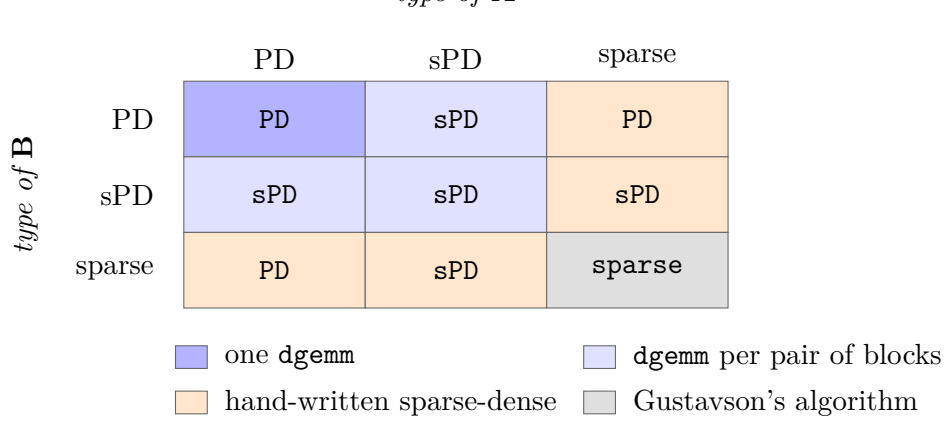


Figure 3.5 Double dispatch of the transposed products $\mathbf{B}^\top\mathbf{A}$ and $\mathbf{B}^\top\mathbf{DA}$. Each cell names the representation of the result for one combination of operand types (PD is permuted dense, sPD is stacked permuted dense), and its colour indicates how the product is computed.

The nine kernels fall into four families. When both operands are permuted dense, the kernel is a single `dgemm` on the two blocks. When an operand is stacked, the kernel pairs the blocks of the two operands and calls one `dgemm` for each pair with intersecting row sets. When one operand is dense and the other is generic sparse, the kernel is a hand-written product that multiplies the stored nonzeros of the sparse operand against the dense block, without converting either operand. When both operands are generic sparse, the kernel is Gustavson's algorithm [55]. A product with a permuted dense or stacked operand produces a result of the same kind, so once a dense block appears, the dense representations take over.

The product of two permuted dense matrices stays permuted dense because the transposed product expands to

$$\mathbf{B}^\top\mathbf{A} = \mathbf{S}_{c_B}^\top \mathbf{X}_B^\top \underbrace{\left(\mathbf{S}_{r_B}\mathbf{S}_{r_A}^\top\right)}_{\text{index-set intersection}} \mathbf{X}_A \mathbf{S}_{c_A}, \tag{3.2}$$

and $\mathbf{BA}$ expands the same way, with $\mathbf{c}_B$ in place of $\mathbf{r}_B$. The term $\mathbf{S}_{r_B}\mathbf{S}_{r_A}^\top$ keeps the rows that appear in both $\mathbf{r}_B$ and $\mathbf{r}_A$, and the outer selectors place the product at the rows $\mathbf{c}_B$ and the columns $\mathbf{c}_A$, so the result is again one dense block at permuted positions. In practice the symbolic pass intersects the two sorted index vectors, the numeric pass multiplies the two blocks with one `dgemm`, and no selector matrix is formed. The cost of this operation depends

on the sizes of the blocks and not on the global dimensions m and n.

3.3.2 Caching structure across solves

Most of the Jacobians in the expression graph of a problem are fixed between solves, since the Jacobian of an affine atom does not depend on the values of the variables. For the structural atoms, such as `index`, `transpose`, and `sum`, the Jacobian is determined by shapes and index sets alone. The other binary affine atoms place problem data in the Jacobian, whose values change only when a parameter is updated. An atom whose data is constant therefore fills its Jacobian at the first evaluation and never touches it again, and a parameter update recomputes only the subtrees that depend on that parameter. The symbolic pass makes this possible by saving the structure of the problem.

In addition, every buffer is allocated during the symbolic pass, including the work arrays used inside the kernels, so the numeric pass never allocates memory. `add` merges the patterns of its operands with a sorted merge, and the sparse products run the symbolic phase of Gustavson's algorithm [58], which finds the pattern of the product row by row. Finding the pattern is the irregular part of both operations, and the numeric phase only fills the stored pattern with values.

The dense representations follow the same split, with the block layout in place of the sparsity pattern. The permuted dense kernels intersect the index vectors of their operands during the symbolic phase, and the stacked kernels also determine which pairs of blocks have intersecting row sets. The numeric phase gathers the matching rows, multiplies them with `dgemm`, and scatters the products to fixed positions.

Finally, the structural atoms save an index map during the symbolic pass. The Jacobian of `sum` is a sum of groups of rows of the child Jacobian. The symbolic pass records, for each of their nonzeros, the output entry it is added to (Figure 3.6). The numeric pass is a single loop that performs one addition per nonzero, and the choice of axis and of representation affects only how the map is built. Other structural atoms such as `index`, `transpose`, and `kron` save their maps in the same way, with a copy in place of an addition.

3.4 Problem interface

`SparseDiffEngine` collects the objective and constraint trees in one `problem` object, defined in `problem.c`. The Python extension module `sparsediffpy` exposes the atom constructors and the functions that build and evaluate the `problem` object. Everything specific to `CVXPY` is in a converter and a canonicalization backend named `diffengine`. The converter builds

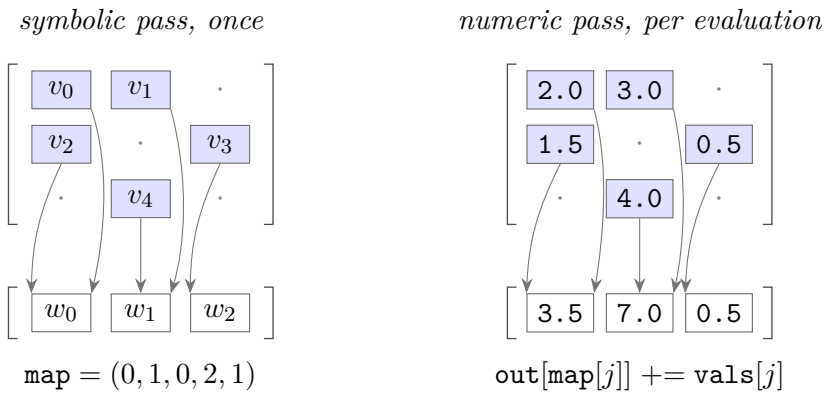


Figure 3.6 The two passes of the `sum` atom for the sum of an expression with three entries. The Jacobian of the sum is the sum of the three rows of the child Jacobian, whose nonzeros v_j are numbered row by row. The symbolic pass computes the pattern of the summed row and the map from each nonzero to the output entry w_i it is added to, shown as arrows. The numeric pass accumulates the values along the fixed arrows.

the equivalent C trees in one pass over the canonicalized problem, with one constructor call per node. The backend obtains the problem data as the values and derivatives of the trees, in place of the matrix stuffing pass of §2.1.

3.4.1 Stacking the constraint Jacobians

The `problem` object stores the constraint trees in the order in which they are registered, and the row blocks of the constraint Jacobian follow that order. Conic solvers expect the constraints grouped and ordered by cone, but the engine knows nothing about cones. Registering the trees in the right order and recording the cone of each row block are both handled by `CVXPY`.

The split between structure and values of §3.3.2 is applied at the problem level as well. The solver receives one global CSR matrix, which the symbolic pass builds once by stacking the sparsity patterns of the constraint Jacobians (Figure 3.7). The numeric pass copies the values of each tree into a fixed slice of the global value array, and skips an affine tree after the first evaluation, because its values cannot change until a parameter does.

The Hessian follows the same split. An affine tree contributes nothing to it, so for a canonicalized problem only the objective is evaluated. The matrix is symmetric, and only its lower triangle is passed to the solver.

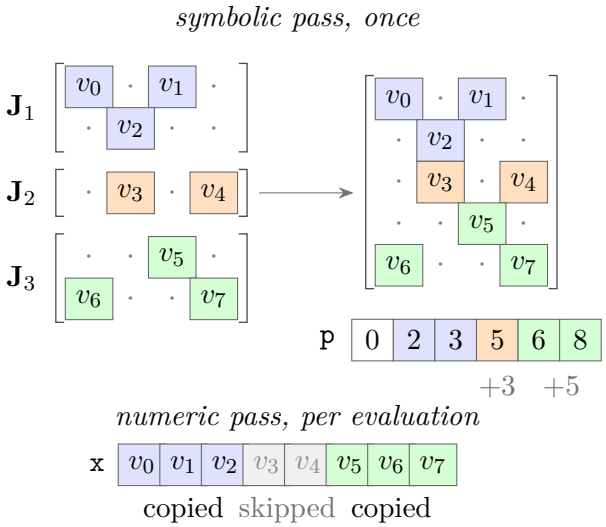


Figure 3.7 The Jacobians of three constraint trees stack into the global constraint Jacobian. The row pointer array `p` concatenates the pointers of the trees, shifted by the nonzeros of the blocks before them. The numeric pass copies the values of each tree into its slice of the value array `x`; the second tree is affine, so the copy is skipped after the first evaluation.

3.4.2 `CVXPY` interface

The converter maps a variable leaf to its column offset in the stacked variable vector, and a parameter leaf to its offset in the flattened parameter vector $\boldsymbol{\theta}$. Each constraint contributes one tree per argument, in the order in which `CVXPY` groups the cones: zero, nonnegative, second-order, semidefinite, exponential, and power.

The `diffengine` backend wraps the `problem` object in one class, which has the same interface as the output of matrix stuffing, so the rest of `CVXPY` is unchanged.

```
class DiffengineConeProgram:
    """minimize c'x + d + (1/2) x'Px subject to Ax + b in K"""

    C_problem             # the problem object and its buffers
    c, d, A, b, P         # problem data, updated in place
    constraints, cone_dims # cone info

    def __init__(self, cvxpy_problem): ...
    def apply_parameters(self, id_to_param_value): ...
    def split_solution(self, solution): ...
```

Listing 3.2 A simplified version of the `DiffengineConeProgram` class.

The class is built once per problem family. Its constructor converts the problem, runs the symbolic pass, and evaluates the `problem` object once, at the origin, which recovers the problem data of (1.2). `CVXPY` applies the sign convention of the cone form afterwards, by negating $\mathbf{A}$.

```
problem.init_derivatives()           # symbolic pass, once
A.rows, A.cols = problem.jacobian_sparsity()
P.rows, P.cols = problem.hessian_sparsity()

x0 = np.zeros(n_vars)                # numeric pass, at the origin
d = problem.objective_forward(x0)    # objective constant
c = problem.gradient()               # objective gradient
b = problem.constraint_forward(x0)   # constraint constants
A.vals = problem.jacobian_values()   # fill the fixed patterns
P.vals = problem.hessian_values()
```

Listing 3.3 A simplified version of the evaluations at the origin that recover the problem data.

After the solve, `split_solution` slices the solution of the stacked variable vector back into the variables of the original problem.

A parameter update only involves the numeric pass. The method `apply_parameters` writes the new values into the parameter leaves and repeats the evaluations at the origin.

CHAPTER 4 BENCHMARK RESULTS

This chapter compares the `diffengine` backend of Chapter 3 against `CVXPY`'s three canonicalization backends: the default `cvxcore` backend of §2.1, denoted `CPP`, and the two pure-Python backends `SCIPY` and `COO`. The comparison is restricted to `CVXPY`'s own backends because `diffengine` is a drop-in replacement for them, and the relevant question is whether it constructs the same problem data faster. All measurements use the 25 problems of `CVXPY`'s public benchmark suite [84], which range from LPs and QPs close to standard form to portfolio, regression, and semidefinite models. Seventeen problems have constant data; the other eight contain parameters.

4.1 Experimental setup

Every benchmark measures the construction of the problem data (P, c, A, b); no solver is invoked. The timed region is a single call to `get_problem_data` with the target solver fixed to `Clarabel` [23]. The `SCIPY` and `COO` backends are selected through the `canon_backend` argument of that call, and `diffengine` by passing `ignore_dpp=True`.

On the parametric problems the default path and `diffengine` build different objects. The DPP path builds the parameter-to-data tensor of §2.1.3, while `diffengine` emits the concrete matrices for the current parameter values. `CVXPY`'s backends are therefore measured twice on these problems, once on the default DPP path and once with `ignore_dpp=True`, on which each parameter is substituted by its value and the backend builds the same matrices `diffengine` does.

`CVXPY` also does not always use the backend it is asked for. When no `canon_backend` is given and the total parameter size reaches 1000 entries, it silently substitutes `COO` for `cvxcore`. The `CPP` columns are therefore obtained by passing `canon_backend="CPP"` explicitly.

All measurements run single-threaded on a MacBook Pro (15-inch, 2018) with a 6-core Intel Core i7-8850H and 16 GB of RAM. Each cold construction is measured in a fresh process, and a measurement is abandoned after 2400 s or when the process is killed by the operating system. Entries are medians of three replications unless a table marks them otherwise. `CPP`, `SCIPY`, and `COO` require unmodified `CVXPY`, so their measurements are taken in a second environment running `CVXPY` 1.9.2 as released, while `diffengine` runs on the fork that carries it. The upstream environment is slightly faster on control problems, so every comparison that crosses the two works against `diffengine`. The benchmark scripts and all recorded

timings are available in the `canonicalization/` directory of the `Jacobian-accumulation` repository[1].

Speed is summarized in two ways. In the tables, the ratio on each problem is `diffengine`'s time over the fastest of `CVXPY`'s three backends on that problem, so values below 1 mean `diffengine` beats all three, and the final row is the geometric mean of the per-problem ratios. The figures report performance profiles [85]. With $t_{p,s}$ the time of backend s on problem p and n the number of problems, the profile of backend s is the fraction of problems on which it is within a factor τ of the fastest backend,

$$f_s(\tau) = \frac{1}{n}\left|\{\, p : t_{p,s} \le \tau \, \min_b t_{p,b} \,\}\right|, \tag{4.1}$$

the minimum running over all four backends. The same profile is reported for memory, with $t_{p,s}$ replaced by the growth of the process's peak resident memory over the timed call.

4.2 Cold canonicalization

Table 4.1 reports the cold times on the 17 problems without parameters, where all four backends construct the same object. `cvxcore` is frequently not the fastest of `CVXPY`'s backends; `SCIPY` beats it on 14 of the 17 problems. Against `cvxcore` alone the geometric-mean ratio would be 0.20 rather than 0.32, a difference attributable to the choice of baseline rather than to `diffengine`.

`diffengine` is the fastest of the four on 14 of the 17 problems, with a geometric-mean ratio of 0.32, that is, 3.1× faster than an oracle that picks the best `CVXPY` backend for each problem. The margins are largest where computing coefficients dominates the construction; UnconstrainedQP, a small problem wrapped in a large `kron` expression, is the extreme case at 121×. The three exceptions, CVaR, OptimalAdvertising, and SDPSegfault1132, are the problems whose construction is dominated by moving one large block of constant data into the problem matrices. Where there are no coefficients to compute, structural knowledge of the atoms buys nothing, and the comparison reduces to how efficiently each backend copies memory.

Table 4.2 reports the cold times on the 8 parametric problems, each `CVXPY` backend measured on both paths. The gap between a backend's two columns is the cost of the tensor. Building it costs `cvxcore` and `SCIPY` 18–58× their direct construction time on the two matrix-stuffing problems with tens of thousands of parameter entries, and neither completes it on the two

[1] `https://github.com/SparseDifferentiation/Jacobian-accumulation`

Table 4.1 Cold canonicalization time in seconds on the 17 non-parametric problems. Bold marks the fastest of the four backends on each problem. The ratio is `diffengine` over the best of `CVXPY`'s three backends; the final row is the geometric mean of the per-problem ratios.

Benchmark	`CPP` (s)	`SCIPY` (s)	`COO` (s)	`diffengine` (s)	Ratio
UnconstrainedQP	10.32	8.52	13.54	**0.07**	0.01
LeastSquares	9.61	10.08	10.68	**1.45**	0.15
Yitzhaki	3.56	3.28	3.55	**0.51**	0.16
HuberRegression	12.05	10.72	11.60	**1.73**	0.16
SimpleQP	11.27	10.59	11.97	**1.87**	0.18
SimpleLP	37.57	29.99	30.96	**6.12**	0.20
TvInpainting	3.89	3.50	3.57	**1.20**	0.34
Cajas	22.67	2.91	2.89	**1.07**	0.37
SlowPruning	7.69	7.82	8.60	**3.01**	0.39
SmallMatrixStuffing	6.97	7.09	10.84	**2.76**	0.40
QuantumHilbertMatrix	4.27	4.05	4.75	**1.67**	0.41
ConeMatrixStuffing	11.06	9.34	16.99	**3.95**	0.42
SemidefiniteProgramming	5.96	2.43	3.78	**2.02**	0.83
Murray	7.13	3.87	4.00	**3.72**	0.96
CVaR	35.85	**24.79**	26.86	24.94	1.01
OptimalAdvertising	6.43	2.32	**2.25**	2.36	1.05
SDPSegfault1132	64.84	**9.51**	43.00	13.21	1.39
Geometric mean					**0.32**

problems with more than 10^6 parameter entries. Only `COO`, whose coordinate-format tensor scales with the number of stored nonzeros rather than with the parameter dimension (§1.4.2), builds the tensor at no extra cost; this degradation is what the substitution rule of §4.1 avoids.

Table 4.2 Cold construction time in seconds on the parametric problems. For each `CVXPY` backend the DPP column is the default path, which also builds the parameter-to-data tensor; the no-DPP column runs with `ignore_dpp=True`, so that all backends produce the same non-parametric matrices. Bold marks the fastest of the four backends on each problem, `diffengine` against the no-DPP columns; the ratio is `diffengine` over the best no-DPP backend. Entries are medians of three replications except where marked; a dash marks a measurement that did not complete.

	CPP		SCIPY		COO			
Benchmark	DPP	no DPP	DPP	no DPP	DPP	no DPP	`diffengine`	Ratio
FactorCovarianceModel	5.12	5.32	4.25	4.32	4.76	4.67	**0.87**	0.20
SVMWithL1Regularization	11.13	11.21	8.21	8.31	9.63	9.79	**1.87**	0.22
SimpleScalarParametrizedLP	4.68	4.56	4.15	4.12	4.49	4.25	**1.03**	0.25
SimpleFullyParametrizedLP†	—	2.29	—	2.16	2.17	2.22	**0.54**	0.25
ParametrizedQP†	—	11.89	—	9.68	10.87	10.76	**2.53**	0.26
ParamSmallMatrixStuffing	7.83	0.28	29.20	0.50	0.40	0.49	**0.21**	0.76
ParamConeMatrixStuffing	3.18	0.18	7.69	0.32	0.27	0.32	**0.14**	0.79
ConvexPlasticity	0.36	**0.35**	0.36	**0.35**	0.35	**0.35**	2.07	5.97
Geometric mean								**0.48**

† Single measurement; all other rows are medians of three replications.

Against the no-DPP columns `diffengine` is fastest on seven of the eight problems, with a geometric mean of 0.48, that is, 2.1× faster than the best `CVXPY` backend. ConvexPlasticity is the one problem in the suite where `diffengine` is clearly beaten. The cause is a sparse materialization defect rather than derivative computation; it is discussed in §5.2.

Figure 4.1 reads the same cold data across the whole suite. `diffengine` is the fastest backend on 21 of the 25 problems, within 2× of the best on 23, and within 7× on all. No `CVXPY` backend is fastest on more than two problems, each needs $\tau \approx 8$ to cover 88–96% of the suite, and each has a tail past $\tau = 100$ on the `kron` problem of Table 4.1.

Figure 4.2 reports the same profile for peak memory. `SCIPY` is the leanest backend, using the least memory on 15 of the 24 problems. `diffengine` is leanest on 7, matches `cvxcore` and `COO` within a few percent in geometric mean, and uses about half again as much memory as `SCIPY`. Its tail is again the problems dominated by one large block of constant data; on OptimalAdvertising its memory grows by 5.1 GB against 0.14 GB for the leanest backend, and on CVaR by 4.6 GB against 0.33 GB. At the other end, on UnconstrainedQP `diffengine`

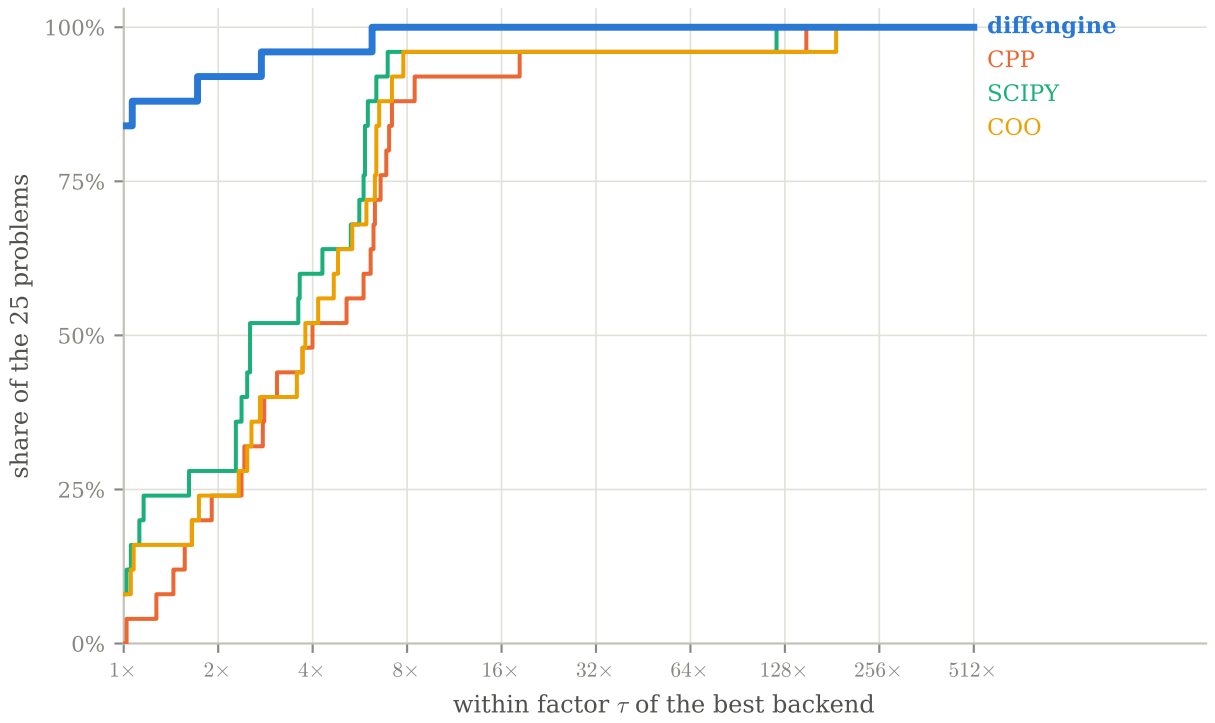


Figure 4.1 Dolan–Moré performance profile (4.1) of cold canonicalization time over all 25 problems, every backend on the `ignore_dpp` path.

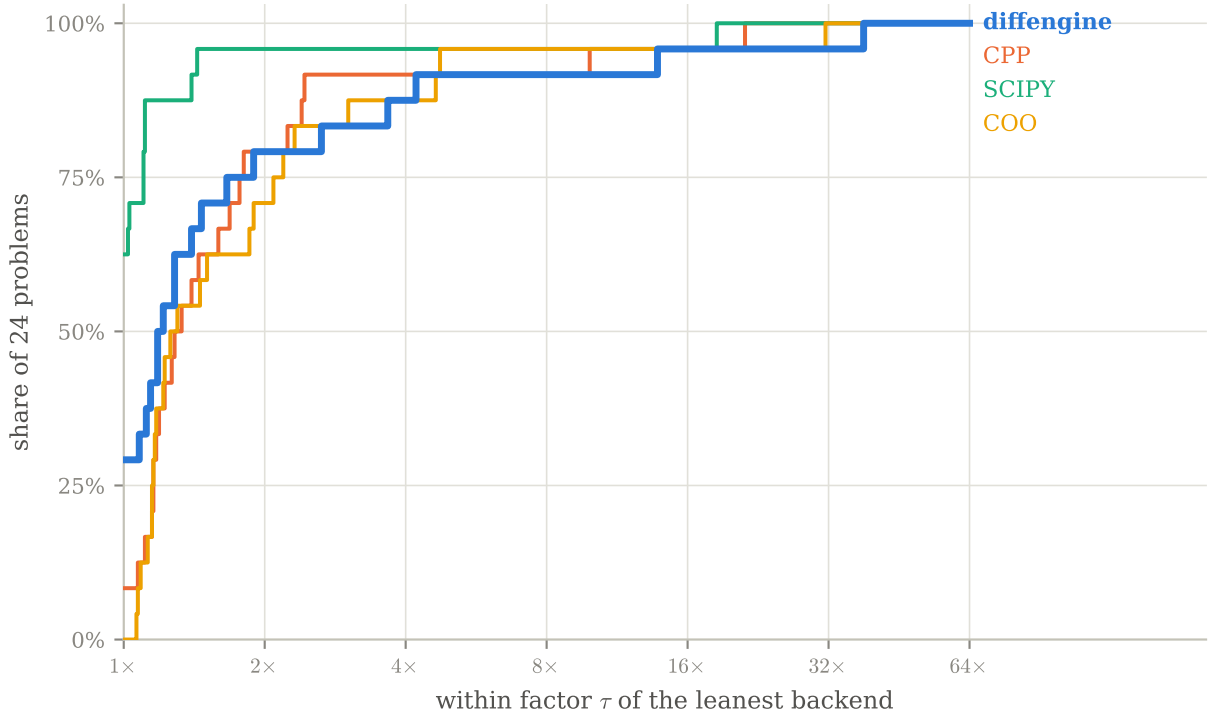


Figure 4.2 Dolan–Moré profile (4.1) of peak memory during cold canonicalization over 24 problems, every backend on the `ignore_dpp` path. ConvexPlasticity is excluded; the memory growth of `CVXPY`'s backends there falls below what the measurement resolves, so no ratio exists.

uses 18× less memory than any `CVXPY` backend. On the excluded ConvexPlasticity its growth of 1.2 GB stems from the materialization defect noted above.

4.3 Parametric re-solve

A parametric problem is re-solved as its parameter values change. After a parameter update `CVXPY` re-applies the cached parameter-to-data tensor, while `diffengine` re-runs the numeric pass on its cached program. Table 4.3 compares the two on the seven problems with a tensor to re-apply; ConvexPlasticity is not DPP as the solving chain evaluates it, so `CVXPY` rebuilds it from scratch on every re-solve.

Table 4.3 Re-solve time in seconds after a parameter update. The tensor column re-applies `CVXPY`'s cached parameter-to-data tensor; `diffengine` re-runs the numeric pass on its cached program. Each entry is the mean of three re-solves; bold marks the faster of the two, the ratio is `diffengine` over the tensor, and the final row is the geometric mean of the ratios.

Benchmark	tensor	`diffengine`	Ratio
FactorCovarianceModel	**0.141**	0.287	2.03
SVMWithL1Regularization	**0.254**	0.496	1.96
SimpleScalarParametrizedLP	0.304	**0.249**	0.82
SimpleFullyParametrizedLP†	0.261	**0.126**	0.48
ParametrizedQP†	1.485	**0.985**	0.66
ParamSmallMatrixStuffing	0.005	**0.001**	0.29
ParamConeMatrixStuffing	0.003	**0.001**	0.29
Geometric mean			**0.71**

† Single measurement; all other rows are medians of three replications.

`diffengine` re-solves faster on five of the seven problems, with a geometric mean of 0.71. The tensor is about twice as fast on FactorCovarianceModel and SVMWithL1-Regularization, the two problems whose parameters enter as small dense maps. On SimpleFullyParametrizedLP and ParametrizedQP the tensor is the one `COO` builds, the only backend that completes it, and `diffengine` is faster still.

4.4 Summary

On cold canonicalization `diffengine` beats an oracle that picks the best `CVXPY` backend for each problem by 3.1× on the non-parametric half of the suite and by 2.1× on the parametric half. Its losses are confined to the problems whose construction reduces to copying one

large block of constant data and to the materialization defect of ConvexPlasticity. Peak memory is at parity with `cvxcore` and `COO` and above `SCIPY`, heaviest on those same problems.

On re-solves the comparison is closer. The cached numeric pass beats re-applying the parameter tensor by 1.4× in geometric mean, and the tensor keeps a 2× advantage where the parameters enter as small dense maps. The tensor must first be built, however, and on the largest parametric problems `COO` is the only `CVXPY` backend that manages to do so.

CHAPTER 5 CONCLUSION

5.1 Summary of Works

5.2 Limitations

5.3 Future Research

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